Artificial Intelligence Review (2026) 59:75
https://doi.org/10.1007/s10462-025-11467-5

# Aigc-driven human-machine intelligence in ITS: technologies, applications, evaluation framework, challenges, and future directions

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Received: 27 May 2025 / Accepted: 8 December 2025
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**Abstract**
This paper explores the integration of Artificial Intelligence Generated Content (AIGC), a rapidly evolving branch of generative AI, with Human-Machine intelligence (HMI) to enhance the functionality of Intelligent Transportation Systems (ITS). As transportation systems grow increasingly complex, adaptive decision-making becomes essential for interpreting vast streams of real-time data from vehicles, infrastructure, and users. AIGC plays a transformative role in optimizing traffic flow through dynamic routing and real-time traffic management, while human intelligence ensures these systems remain responsive to evolving real-world conditions. For safety, AIGC is used to simulate complex driving scenarios for autonomous vehicle training and detect traffic anomalies, with human oversight providing contextual decisions in ambiguous situations. For sustainability, AIGC supports data-driven strategies to reduce emissions and energy use, while human expertise ensures alignment with ethical and environmental goals. This synergy enhances real-time decision-making, improving both accuracy and adaptability across ITS scenarios. The paper presents a comprehensive review of core and supporting AIGC technologies and their applications across key ITS domains. Case studies and initiatives from industry leaders demonstrate practical implementations of AIGC-driven HMI collaboration. To guide future deployments, we propose a conceptual five-layer evaluation framework for assessing AIGC-HMI systems, encompassing functional performance, human interaction, explainability, ethical compliance, and robustness. We also address challenges such as legacy system integration, data privacy, model bias, and scalability. The paper concludes by outlining future research directions, emphasizing the need for scalable, interpretable, and ethically aligned AIGC models. This work contributes to the development of intelligent, adaptive, and trustworthy transportation systems.

**Keywords** Artificial intelligence · Artificial intelligence generated content · Generative artificial intelligence · Human-machine intelligence · Intelligent transportation systems

Extended author information available on the last page of the article

Published online: 05 January 2026

## 1 Introduction

Intelligent Transportation Systems (ITS) have been considered as a basis of future transportation solution with an emphasis on solving urban traffic problems related to traffic congestion, road safety, and environmental friendliness Hossain et al. (2024a). Utilizing sensing, communication and computing intelligence, ITS enhances the operating efficiency and the user experience, and reduces the energy consumption Khalil et al. (2024). Yet, as an evolving, data-rich environment with continuous massive real-time output from vehicles, infrastructure, and road users, making traditional rule-based or static AI approaches to under-perform in their ability to adapt. This gap calls for new generative capabilities and adaptive decisions, in which Artificial Intelligence Generated Content (AIGC) is the promising solution.

AIGC is a generative artificial intelligence (GenAI) that can autonomously generate content across a multitude of media including text, image, video, audio, 3D, and multi-modal forms of expression Zhang et al. (2024b); Guo et al. (2024). The ITS field takes advantage of AIGC in the production of dynamic scenario simulations (Kim et al. 2021), traffic prediction Ren (2024), route guidance Lin et al. (2024a), and emergency response strategies (Xu et al. 2024f). Whereas outputs from standard AI processing are pulled from a repository and cannot be used for forward-looking analyses, AIGC systems construct new conclusions and outcomes on the fly, providing predictive and generative intelligence to adaptively manage transportation operations Li et al. (2025).

The AIGC methods are not adequate to address multiple intricacies in the transportation systems of the real world. The AIGC system is suitable to be realized based on Human-Machine intelligence (HMI) integration among other factors for contextual understanding, ethical decision-making, and adaptability according to Yang (2024). AIGC tracks traffic congestion trends to offer traffic rerouting alerts, but there is also human operator that maintain essential situational awareness that allow them to adjust strategies on the fly in response to traffic closures, construction zones, and other incidents. Human judgment is always necessary for AVs decision-making to deal with the uncertain situation, such as various considerations for public safety in ethical dilemmas or new weather conditions beyond the interpretation capability of algorithms or both Zhang et al. (2024b). AIGC may create effective public transit schedules or emergency response plans, but these to work, need human involvement to make them useful for application, as they must adhere to community needs, cultural norms, and operational restrictions. The synergy of AIGC and HMI generates innovative ITSs that are technically robust, socially grounded, responsive, and ethically responsible. One of the strengths of ITS arises from the synergy between human and AIGC as humans provides contextual awareness, ethical implications, and operational resilience, which complements the quick nature of data-driven solutions of AIGC.

Integration is promising but brings challenges of system compatibility within existing infrastructure, privacy concerns, model bias, and scalability, as reported by Papandreou (2024). The realization of the AIGC-driven HMI depends on the satisfaction of explainability, fairness, and user-specific adaptability in a real-time manner for safe and effective processing. To tackle these challenges a cross-disciplinary approach, bridging technological advances with domain knowledge, ethical principles and regulatory frameworks, is needed.

Accordingly, in this paper, we formulate a holistic and systematic review on AIGC-driven HMI in ITS. It combines technology landscape analysis with reviews of implementations in

practice to provide practical tools and advice on the development of future AIGC-empowered HMI in ITS.The major contributions of this paper are summarized as follows:

1. Explores AIGC technologies and their applications across key ITS domains, including traffic management, safety, diagnostics, environment, and smart cities.
2. Demonstrates real-world implementations of AIGC-driven HMI systems through case studies from industry leaders, validating their effectiveness in enhancing real-time decision-making and operational resilience.
3. Proposes a novel five-layer conceptual evaluation framework to assess AIGC-HMI systems in ITS, addressing the current gap in systematic performance, explainability, and human interaction assessment.
4. Identifies key challenges to AIGC-driven HMI adoption, such as legacy integration, privacy, and bias, and presents actionable strategies for ethical, scalable deployment in transportation networks.
5. Outlines future research opportunities to advance AIGC-driven HMI for ITS, focusing on multimodal generation, adaptive learning, and human-in-the-loop collaboration.

### 1.1 Research trends in AIGC for ITS

In order to explore the development and changing priorities of AIGC research in ITS, a year-wise analysis of publications during 2018 to October 2025 was carried out. This research studies six important generative modalities that are pertinent to ITS: text, image, video, audio, 3D and multimodal generation. Data were retrieved from the Scopus database in October 2025 using targeted search terms including “AIGC,” “generative AI,” “text generation,” “image synthesis,” “video generation,” “audio synthesis,” “3D scene generation,” “3D content synthesis,” “multimodal generation,” and “intelligent transportation systems.” The research included only English peer-reviewed journal and conference articles after removing duplicates to maintain accuracy.

The research activity shows a steep and continuous rise according to Fig. 1, especially in image and text generation domains. This increasing trend demonstrates their usefulness for ITS applications, including traffic visualization and automated reporting, and infrastructure communication.

Video generation, which is important for scenario simulation, digital twin building, and AV training, also displays significant growth, indicating higher recognition from both academia and industries’s temporal modeling ability. Meanwhile, although development is in its infancy, multimodal AIGC is growing as an emerging field. There is also a growing interest from the ITS industry in ubiquitous systems, which take visual, textual, and audio inputs and provide human-friendly decision support for the process of decision-making.

Audio generation is a domain that researchers have not yet thoroughly investigated. Further studies should address this opportunity of creating voice-guided navigation, multilingual alert systems, HMI auditory feedback for inclusive, real-time ITS applications.

These trends in publishing indicate a clear bias towards applications in the ITS domain that leverage AIGC to generate more engaging and cooperative, context-aware HMI solutions. The findings of this study indicate that this review is up to date, and audio and multimodal integration are key areas of research for future studies.

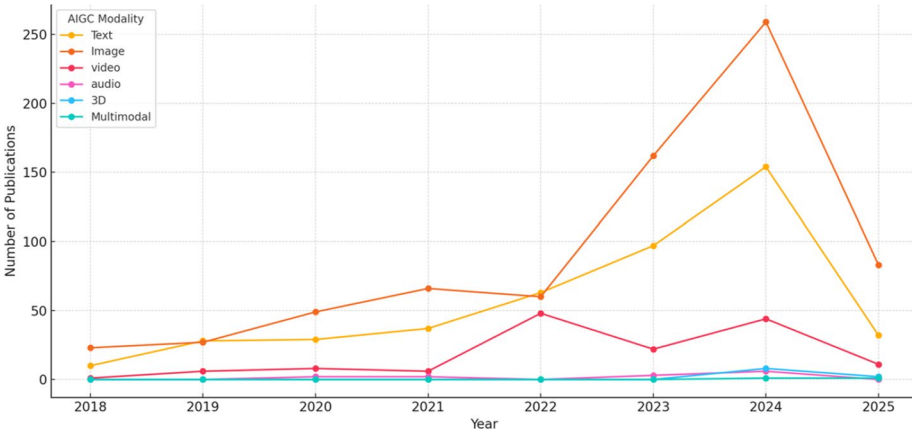



**Fig. 1** Year-over-year publication trends (2018–2025) across AIGC modalities applied to ITS, based on Scopus data as of October 2025

**Table 1** Distribution of research studies used in this review

| Database source | No. of research studies |
|---|---|
| IEEE Xplore | 78 |
| arXiv | 25 |
| Science Direct | 21 |
| Springer | 23 |
| MDPI | 10 |
| ACM Digital Library | 7 |
| Taylor & Francis | 5 |
| Wiley | 5 |
| Other sources | 68 |
| Total | 242 |

## 1.2 Research methodology

The review follows a structured literature review methodology using the PRISMA guideline for transparent systematic analysis Haddaway et al. (2022). The PRISMA flow diagram offers a structured selection process including five stages: identification, screening, relevance, refinement, and inclusion, which assure the reliability and validity, and reproducibility of this review.

The identification stage begins with a thorough literature search across eight major scholarly databases, which include IEEE Xplore, arXiv, ScienceDirect, SpringerLink, MDPI, ACM Digital Library, Taylor & Francis, Wiley, and additional relevant sources, including websites and industry reports as illustrated in Table 1. The search included publications from 2020 through October 2025. The search strings used to find relevant articles included (”AI-generated content” OR ”AIGC” OR ”Generative AI”) AND (”Human-Machine intelligence” OR ”Human-Machine collaboration” OR ”human interaction”) AND (”intelligent transportation systems” OR ”autonomous driving” OR ”traffic management”). The search produced 1,950 publications, which are displayed in Fig. 2.

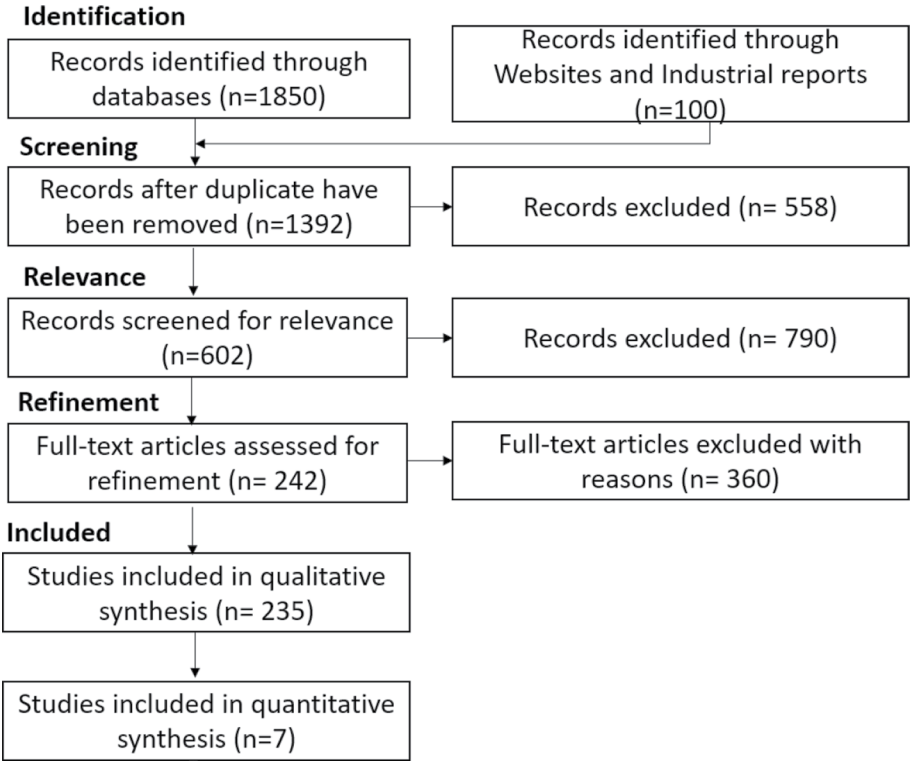



**Fig. 2** The PRISMA flowchart

The screening stage involved duplicate removal followed by application of predefined inclusion criteria to select studies. This research selected original studies published in journals and arXiv, and conference proceedings that were written in English. The studies needed to directly discuss AIGC, HMI, and ITS. The screening process eliminated 558 papers because they were duplicates, or their full text was restricted, or they were written in languages other than English.

The relevance stage required researchers to assess abstracts and titles to determine the direct connection between studies and AIGC and HMI in ITS. The review process focused exclusively on studies that directly investigated AIGC and HMI applications in ITS. This process removed 790 studies because they did not fulfill the requirements for AIGC and HMI, and ITS relevance.

The stage of refinement required that the full text of the remaining articles be assessed to judge their quality in terms of methodology, relevance to the domain, and availability. A study not meeting quality standards in its methodology, or only weakly related to AIGC-driven HMI in ITS is the subject matter of the paper to be excluded. The researchers excluded 360 studies because they had inadequate experimental design or data analysis, or did not focus on AIGC and HMI, and ITS fusion.

The inclusion stage allowed researchers to choose studies that met all the inclusion criteria. The selection of studies was based on their methodological soundness and direct relevance to AIGC, HMI in ITS. A total of 242 studies were included in the final analysis,

providing a comprehensive and balanced review of the current state of AIGC-driven HMI systems in ITS, covering their technologies, applications, challenges, and future research directions.

### 1.3 Related surveys

Research has been conducted to understand the increasing use of Generative AI and its implementation in transportation systems. There is a lack of surveys that examine the relationship between AIGC and HMI in ITS. Table 2 presents a comparative analysis of related surveys that differ in their focus domains, modalities, human-AI collaboration and intelligence, ITS relevance, and inclusion of case studies.

Yan and Li (2023) presents a comprehensive review of generative AI methods used in ITS by dividing their work into traffic perception and prediction and simulation and decision-making categories. The authors provide comprehensive information about fundamental ITS operations, but they fail to discuss Human-Machine collaboration and evaluation methods, which are vital for high-risk ITS applications.

**Table 2** Comparison of related surveys on AIGC, HMI for ITS

| Survey | Focus domain | Modalities covered | HMI | ITS-specific | Case study |
|---|---|---|---|---|---|
| Yan and Li (2023) | GenAI for ITS | Text, Image, Video, Multimodal | No | Yes | No |
| Shi et al. (2023) | Human-GenAI Interaction | Text, Image, Audio, 3D | Yes (HMI taxonomy) | No | No |
| Ren (2024) | AIGC for Urban Intelligent Traffic Management | No | No | Yes | No |
| Fu et al. (2024) | PLMs in Urban Science | Text | Yes (on text only) | No | No |
| Xu et al. (2024b) | GenAI + Urban Mobility | Text, Image, 3D | No | Partially (Smart cities) | No |
| Liu et al. (2024a) | GenAI for Unmanned Vehicle Swarms | Text, Image, Video | Yes (Partially) | Partially (AVs) | No |
| Rong et al. (2025) | AIGC for ITS Subsystems | Text, Image, Audio, Multimodal | Mentioned briefly | Yes | No |
| This survey | AIGC + HMI for ITS | Text, Image, Video, Audio, 3D, Multimodal | Yes (core focus on HMI) | Yes | Yes |

The research by Shi et al. (2023) presents a human-computer interaction (HCI)-centric survey that develops a six-dimensional taxonomy for human-GenAI interactions. Their framework for feedback mechanisms, control levels, and engagement establishes essential principles for understanding how generative systems and human oversight can coexist, which this paper expands in ITS contexts through specific AIGC-driven HMI examples. Ren (2024) studies AIGC application scenarios in urban traffic control and management for better system efficacy and adaptivity. This study does not provide a comprehensive evaluation of the AIGC modalities and also the Human-Machine collaboration and multimodal integration, which is what our survey focuses on. Fu et al. (2024) studies on uthe tilization of the implementation of pre-trained language models (PLMs) in urban science. The authors highlight applications including disaster sensing and behavior modeling, as well as the challenges of human-AI collaboration. The authors' analysis, therefore, is only concentrated on text-based PLMs and does not consider multimodal AIGC or those specific to ITS. Xu et al. (2024b) ccarriesout a scoping review on generative AI in urban digital twins with an emphasis on 3D content generation for planning of the built environment. The paper is applicable to urban mobility but not specific to ITS and does not consider HMI techniques, or deployment maturity. The study by (Liu et al. 2024a) offers a review on generative AI applied to unmanned vehicle swarms in a domain-oriented manner. The topics of dynamic coordination and generative communication have also been studied by the authors, with real-time decision-making. Their work, however, neglects the Human-Machine collaboration, system assessment, and ethical issues for large-scale implementations of ITS. Rong et al. (2025) analyzes systematically the AIGC for each subsystem of ITS, i.e., road, vehicle, traveler, and management in ITS. The authors underline key features such as dialogue, reasoning and multimodal content generation. While the study offers a broad range of coverage of applications, it does not consider the human-in-the-loop, the interpretability and deployment metrics. Our survey builds upon their work by considering HMI models and evaluation methodologies, as well as multimodal integration approaches to deploy practical systems.

The reviews to date focused on AIGC technologies only or ignored the aspect of collaboration with humans, or were not providing a complete multimodal and evaluative analysisi. To fill the existing gaps, this review presents for the first time, a comprehensive survey of AIGC modalities with HMI techniques, structured performance evaluation, and ITS-specific deployment practices.

### 1.4 Paper organization

The rest of the paper is organized as follows: Since they play crucial roles in ITS systems, AIGC and Human-Machine intelligence are briefly introduced in Sect. 2. The key enabling technology and supporting technique of AIGC-based HMI in ITS is elaborated in Sect. 3. The applicability of AIGC-driven HMI systems to various ITS fields is discussed in Sect. 4. Case studies and initiatives that relate to the generation of practical impacts by AIGC are presented in Sect. 5. A conceptual five-layer evaluation framework for AIGC-driven HMI systems is formulated in Sect. 6. In Sect. 7, we discuss the challenges towards the integration of AIGC in ITS and potential solutions. Sect. 8 concludes with a discussion on promising future research directions and industry applications to further promote the use of AIGC in transportation. Finally, in Sect. 9, the paper concludes with a summary of findings and their implications for the future of AIGC.

## 2 AIGC and human-machine intelligence: an overview

### 2.1 AIGC overview

AIGC relies on the advanced generative AI models to generate, simulate, and optimize content in a wide range of applications, including education Li et al. (2024a) (production of the customized learning contents), healthcare Shao et al. (2024) (facilitate research and treatment), entertainment and gaming Blanchfield (2023) (novel immersive experiences), finance Wu et al. (2023a) (enhance data analysis and customer services), and tourism Wong et al. (2023) (promoting autonomous travel planning and decision-making). Rather than traditional AI approaches that generally rely on explicitly given rules or supervised learning, AIGC is based on generative models that have the capability to generate contents which are largely indistinguishable from human-generated outputs Mohamadi et al. (2023). For example, GPT-4, LLM, can produce coherent, context-aware text, and DMs such as DALL-E and Stable Diffusion can generate photorealistic images from text prompts Bengesi et al. (2024). Such features are not limited to content generation, but also to decision-making and scenario simulation in real time, which leads to AIGC having a high value for dynamic systems, e.g., ITS.

Within the scope of ITS, AIGC brings great benefits to the traffic management system modeling, prediction, and optimization. For instance, AIGC may be employed to model intricate traffic patterns for city planning, allowing policymakers to evaluate several infrastructure designs at once without incurring real-world costs Lin et al. (2024a). In addition, in scenarios where data are scarce (e.g., night driving or adverse weather conditions) Zhang et al. (2024b), AIGC can produce synthetic data for autonomous vehicle algorithms. These are just a few of the reasons why AIGC can be used to increase the efficiency, reliability, and scalability of ITS solutions.

### 2.2 Human-machine intelligence (HMI)

Human-Machine Intelligence (HMI), also known as hybrid or collaborative intelligence, describes the combined power of human thinking with machine (AI) computational abilities to enhance ITS decision-making through adaptive, transparent, and accountable processes Muzahid et al. (2024). Machines perform best at automated work, massive data processing, and predictive analytics, yet humans deliver essential capabilities for contextual understanding and ethical reasoning, creativity, and situational awareness in dynamic or uncertain traffic environments.

The system design in Fig. 3 demonstrates how HMI allows real-time Human-Machine intelligence collaboration through human judgment enhancement of machine efficiency. The system depends on multiple methodologies to achieve effective interaction and mutual learning between human and machine intelligence. The methodologies form the base of ITS collaborative intelligence, which enables AI-powered systems to stay flexible while maintaining ethical alignment and human input responsiveness.

1. Human-in-the-Loop (HITL): This methodology integrates human oversight into AI decision-making processes. The ITS system employs HITL in traffic control centers through which AI generates rerouting strategies that human operators verify or modify

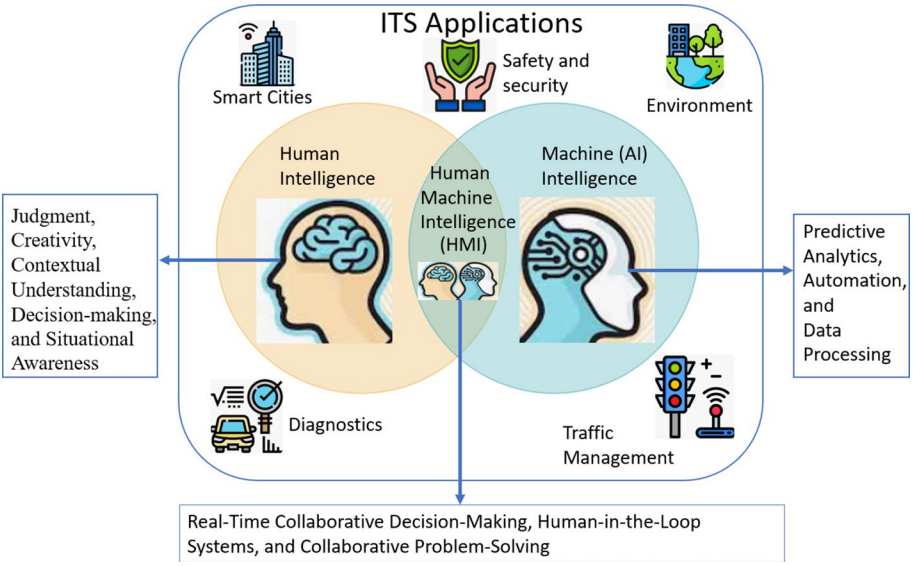



**Fig. 3** Human-machine intelligence framework in ITS

before deployment according to Huang et al. (2024); Tyburzy et al. (2024). HITL allows human operators to make essential decisions through their judgment while maintaining accountability.

2. Active Learning Loops: Active learning depends on ongoing human feedback to enhance machine learning models throughout time. In ITS, human annotators refine traffic incident classifications or correct model outputs during training cycles Maennel and Maennel (2024). Human reviewers identify unrealistic elements in synthetic scenario generation for AVs, which leads to better future simulations. The repeated human feedback loop produces models that become more accurate and reliable.
3. Adjustable Autonomy: This methodology allows machines and humans to share control based on the current situation. The AI system operates lane-keeping and adaptive cruise control functions in semi-AVs, but human operators need to take control during complex situations. The human designer can use a shared knowledge base to guide the AI system while modifying machine-generated car frontal designs according to their cognitive mental model in Human-Machine hybrid design. The system's adaptability allows designers to achieve their creative vision through the hybrid intelligence system described in Wu et al. (2023b).
4. Shared Control Interfaces: This methodology allows for systems outputs to be accessible to both human users and machines. According to Rousseau et al. (2024), AI in traffic management systems forecasts areas of traffic congestion, but human controllers still have the authority over the parameters as well as the possibility of a manual override. Hernández (2021) illustrates how threats are detected and responses are suggested by AI systems, and how human operators are able to modify or decline those suggested responses by means of common control interfaces. Together, human and AI capabilities allow critical decisions to be made via shared control interfaces that leverage AI speed with human operational expertise, a winning solution for AV cyber incident response.

These four approaches are the foundation for ITS collective intelligence. The application of these methods enables AI-driven solutions to be both adaptable and ethically aligned with human decision-making, which provides the benefit of increased public trust and operational dependability in the context of complex transportation systems.

### 2.3 Synergizing AIGC and HMI in ITS

The combination of AIGC with HMI methods results in a dynamic scheme that allows real-time decisions in the ITS environment. AIGC produces complex outputs (traffic simulations, dynamic routing plans, anomaly detection, etc.) and HMI ensures that these outputs remain contextually relevant, ethically sound, and applicable to the dynamic conditions of the real world. It combines the predictive power of AI with human intuition and can be applied to decision making in any critical environment, including traffic management and AV systems.

1. Human-in-the-Loop (HITL): AIGC traffic solutions (e.g., routing policies, signal plans, or congestion predictions) are generated automatically in the first place, and are then subjected to human supervision. Human operators at traffic management centers review AIGC diversion suggestions for traffic, but also take into account road works, local events, and weather conditions, and may modify or overrule those suggestions (see Liang and Li (2024)). Real-time verification affords a degree of accountability and safety precautions, particularly in circumstances where human discretion is still necessary.
2. Active Learning Loops: Active learning loops depend on human feedback to improve machine learning models throughout time. Ruan et al. (2024) present a text-to-traffic scene framework for AV training, which enables users to specify parameters including weather conditions, vehicle types, and road signals. The AIGC system produces multiple traffic scenarios while human feedback maintains both the accuracy and variety of generated data. The repeated process strengthens the AIGC's learning abilities, which results in more realistic and applicable synthetic traffic scenarios for AV training.
3. Adjustable Autonomy: The AIGC in AV systems produces commands to control lane-keeping functions, speed regulation, and obstacle avoidance capabilities. AI takes complete control during normal system operation. The system transfers control to human drivers when it encounters unpredictable situations, including unpredictable pedestrian movements or sensor malfunctions, or unorganized intersections, according to Lv (2023). The system allows for dynamic allocation of control, where the machine performs the standard operations and the human takes over when complexity demands their expertise. The precision of AIGC combined with human intuition leads to a system that guarantees the proper functioning of AV systems.
4. Shared Control Interfaces: Shared Control Interfaces allow human operator and AIGC systems in ITS to collaborate and interact cooperatively with each other. The AIGC system produces traffic congestion predictions by analyzing data, but the predictions can be altered by human operators who are aware of local conditions and know of any public events that might have an effect. The system through this approach of shared control becomes flexible and context-aware. The 51 WORLD Scenario Copilot tool creates synthetic traffic scenarios for AV training, but human input modifies and enhances these

scenarios to more closely mirror real-world complexities, such as erratic traffic events and shifting weather patterns Medium (2023).

These real-time, iterative interactions between AIGC and HMI show that AIGC systems operate as system components contrast to standalone systems. By working alongside human operators, the AIGC systems evolve into more effective decision-making assets that can adapt to scenarios and that uphold ethical principles with contextual awareness. The complementary nature of AIGC and HMI methodologies enables the development of ITS applications that are intelligent and efficient, and that provide explainable systems that are transparent and socially acceptable.

## 3 Key technologies enabling AIGC-driven HMI for ITS

AIGC-driven HMI in ITS rests on two pillars: (i) core generative modalities: text, image, video, audio, and multimodal that create or summarize content for operator support and citizen communication; and (ii) supporting infrastructure: computer vision, IoT, edge/cloud computing, 5 G/6 G connectivity, digital twins, and blockchain that supplies perception, context, latency, and governance. This section first compares the core modalities on capability, latency, and deployment readiness, then analyzes how the supporting stack composes to meet real-time and audit requirements.

### 3.1 Core AIGC technologies

AIGC for ITS spans five complementary content modalities: text, image, video, audio, and multimodal that together support perception-to-action loops for incident summarization, traveler communication, planning/simulation, and operator guidance. Figure 4 situates these modalities with representative tasks.

Within this landscape, mature text generation is driven by transformer language models: GPT, BART, and T5 in particular which turn heterogeneous transportation streams into concise, context-aware narratives used for incident and operations briefs, real-time alerts, adaptive navigation instructions, and public announcements Yenduri et al. (2024); Haruna et al. (2022); Hanif (2023); Qu et al. (2023). Image synthesis relies on diffusion models, GANs, VAEs, and normalizing flows to create high-fidelity traffic scenes for scenario modeling, urban design "what-ifs," and autonomous vehicle perception robustness; promptable stable diffusion class systems help explore congestion hot spots and infrastructure changes, while GAN pipelines target rare or hazardous edge cases Jamal et al. (2024); Lin et al. (2023a); Huang (2024); Coeurdoux et al. (2022); Xu et al. (2024e); Luo et al. (2024b); Jeng (2025). Extending from stills to dynamics, video generation via temporal GANs and video diffusion provides time-consistent clips suited to operator drills, public-safety communication, and bottleneck illustration Munoz et al. (2021); Tulyakov et al. (2018); Xing et al. (2024); Satpathy et al. (2025). On the audio channel, neural text-to-speech: WaveNet, Tacotron, WaveGlow, VALL-E delivers multilingual, accessibility-friendly announcements (detours, closures, station/stop messages) with edge-friendly latency, complementing text where hands-free interaction and inclusivity matter Matsumoto et al. (2022); Aziz et al. (2023); Zhao et al. (2020); Kim et al. (2023); NVIDIA (2025); Zhou et al. (2024a). Finally,

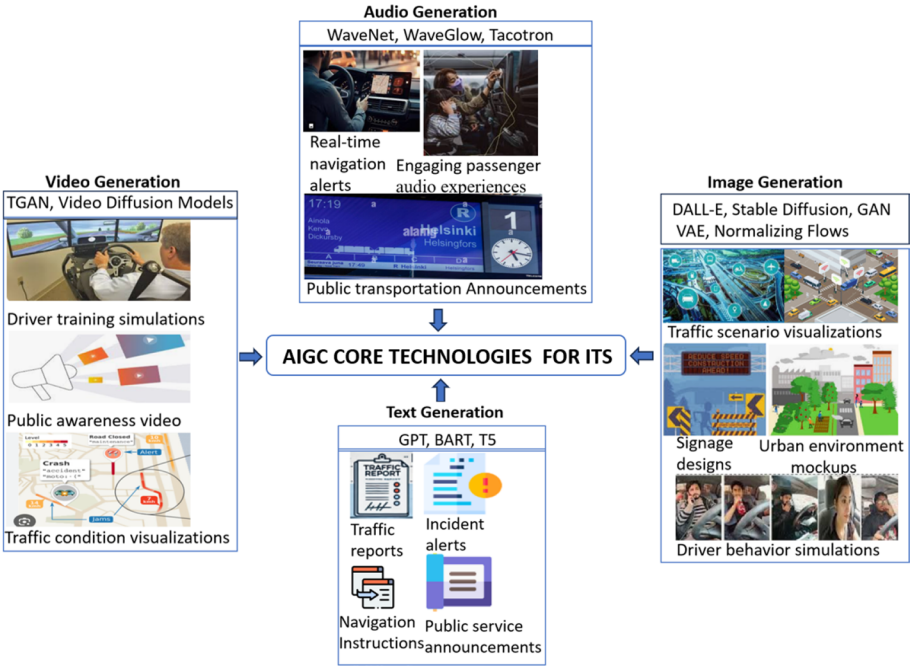



**Fig. 4** Functional roles of core AIGC content modalities in ITS applications

multimodal vision–language models (e.g., GPT-4V/Flamingo-class, BLIP-2/LLaVA-style) align perception with language to produce scene-grounded Q&A, evidence-cited summaries, and coordinated text–audio outputs; recent prototypes fuse visual, acoustic, and sensor streams for hazard detection and accident analysis Rong et al. (2025); Golam et al. (2025); Wu et al. (2024). Figure 5 illustrates how upstream video/sensor understanding conditions downstream text and audio for operator-in-the-loop use.

Practical ITS deployment hinges on controllability, grounding, latency, and governance. Operator-in-the-loop tasks typically require end-to-end p95 (95th-percentile) latency within a few seconds, which favors text and TTS at the edge; image and especially video generation is most effective as precomputed assets served from digital-twin/back-end stores given compute cost and the need for realism/coverage validation Ghosh et al. (2024). Multimodal pipelines orchestrate video/sensor understanding with downstream text and audio (e.g., an incident description plus a voice alert), improving situational awareness while increasing GPU memory (VRAM) and inference latency; selective multimodality and retrieval grounding help keep latency within operational targets.

As summarized in Table 3, the deployment landscape exhibits uneven maturity: text/audio are production-ready for real-time operator workflows, whereas video and multimodal are generally pilot-stage online (but production-grade when precomputed). This asymmetry forces systems to bridge reliable but narrower single-modality capabilities with richer, computationally heavier multimodal solutions.

A cross-modal pattern emerges. Text offers the best speed and coverage for operator workflows but requires retrieval grounding and verification to control hallucinations Ji et al. (2023). Multimodal VLMs materially improve situational awareness by coupling visuals

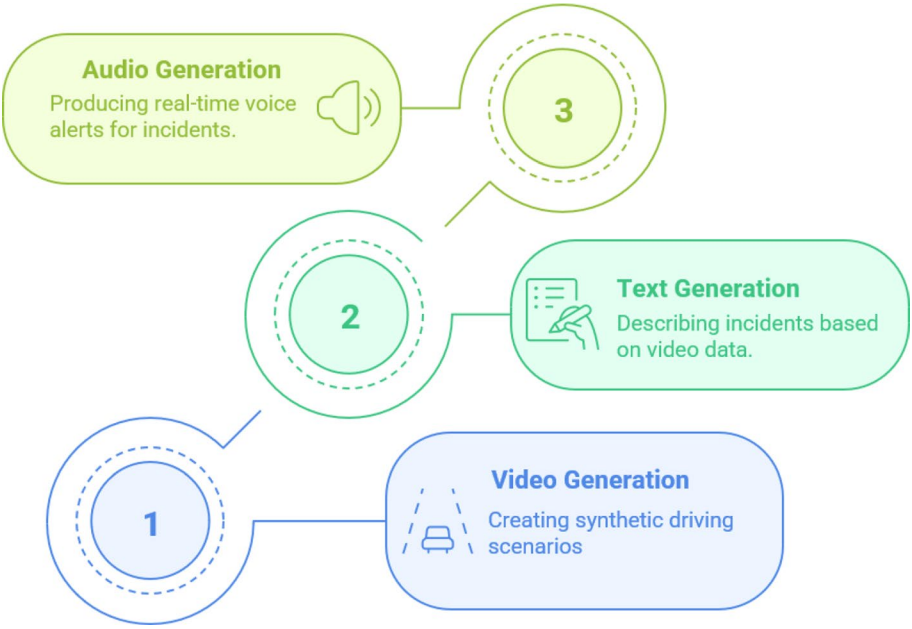



**Fig. 5** Multimodal AIGC workflow in ITS: coordinated generation and interaction across video, text, and audio modalities

and text, at the cost of higher VRAM and seconds-scale latency unless tiered (edge distillation plus a cloud VLM) Zhang et al. (2024f). Image and especially video generation excels at realism and rare-event coverage for offline training and planning, while online use is constrained by compute and controllability Liu et al. (2024b). By contrast, audio/TTS is latency-tolerant and production-ready for edge-served alerts, but speed alone is insufficient; reliability and governance ultimately determine deployability.

Beyond latency, accuracy and reliability are decisive for safety-critical ITS. Ungrounded LLM/VLM outputs can fabricate entities or causal links; retrieval augmentation, provenance-aware prompts, structured outputs, and refusal/verification policies materially reduce critical factual errors. Visual synthesis faces domain gaps when synthetic textures, weather, or occlusions diverge from local distributions; distribution matching, human validation gates, and downstream Key Performance Indicator (KPI) checks (e.g., perception lift on held-out real data) mitigate risk. For audio, there are controls for consent and voice cloning; watermarking, speaker ID limits, and locale-specific QA assist in mitigating misuse and pronunciation variability. Deployed systems pair these mitigations with audit logs and approval workflows, especially where incident communications or operator guidance are involved Zhou et al. (2024c).

Bridging research and operations requires more than benchmark fidelity: production ITS demands robustness to drift, governance (personally identifiable information (PII) redaction and auditability), and disciplined change control (prompt/model versioning). In practice, four levers close this gap: (i) retrieval grounding with freshness service-level agreements (SLAs) so recommendations reflect current Standard Operating Procedure (SOPs) and data; (ii) policy guardrails and clear escalation trees for operator-in-the-loop decisions; (iii) tiered

**Table 3** Comprehensive assessment of core AIGC modalities for ITS

| Modality | Representative models | Key trade-offs & limitations | Compute intensity | Latency (p95) | ITS use cases | Feasibility/ readiness |
|---|---|---|---|---|---|---|
| Text | GPT-4/4o, Llama 2/3, Mistral; T5, BART; RAG-LLMs | **Trade-offs:** accuracy vs speed; generality vs domain specificity. **Limits:** hallucinations if ungrounded; prompt sensitivity; knowledge cutoff; multilingual consistency | Low–Medium (CPU/edge GPU; context size matters) | Sub-second to few seconds (with caching & retrieval) | Incident/ops summaries; SOP lookup; multilingual alerts; operator rationales; policy drafts | **Production** (mature tooling; ROI proven; integrates with RAG/ guardrails) |
| Image | SDXL/SD3; StyleGAN; VAEs; Flow models | **Trade-offs:** controllability vs fidelity; diversity vs domain relevance. **Limits:** domain gap; weather/locale bias; artifacts; style drift | Medium–High (GPU VRAM; tiling/fast samplers help) | Few seconds (online constrained; precompute preferred) | Infrastructure visualization; rare-case synthesis for perception; signage/ streetscape mockups; adverse-condition augmentation | **Research/Pilot** (online); **Production** (precomputed; requires validation pipelines) |
| Video | Video diffusion/latent video; MoCoGAN/TGAN; (e.g., Sora/Gen-2*) | **Trade-offs:** temporal coherence vs compute; realism vs speed. **Limits:** high resource demand; temporal artifacts; storage/ egress overhead | High (often multi-GPU; memory intensive) | Tens of seconds (online impractical; precompute/batch) | Scenario libraries; safety training clips; congestion pattern visualization; DT playback for drills | **Pilot** (batch/offline); **Production** (precomputed libraries; infra cost & QA constraints) |
| Audio | Tacotron/ FastSpeech; WaveNet; VALL-E; HiFi-GAN | **Trade-offs:** voice quality vs speed; prosody control vs cost. **Limits:** consent/voice cloning; accent bias; low-resource locales | Low (edge-deployable; quantization effective) | Sub-second (edge-optimized; cacheable) | Real-time detours & warnings; station/stop announcements; accessibility HMI; hands-free operator prompts | **Production** (edge latency proven; mature integration; clear regulatory patterns) |
| Multimodal | GPT-4V class; LLaVA/ IDEFICS/ BLIP-2; Q-Former hybrids | **Trade-offs:** versatility vs specialization; capability breadth vs latency/VRAM. **Limits:** higher VRAM; OCR/perception failures; prompt complexity; image privacy | Medium–High (GPU; quantization + retrieval help) | Few seconds (tiered edge–cloud + caching) | CCTV snapshot triage & explanation; incident verification; infrastructure inspection notes; multi-sensor interpretation | **Pilot/Limited Ops**; **Production** (with strict guardrails: privacy, audit, approval-by-exception) |

*Performance context.* Operator-in-the-loop deployments typically target end-to-end p95 (95th percentile) latency within a few seconds (ingestion → redaction → retrieval → generation → verification). *Targets vary by task* e.g., sub-second for TTS alerts vs. seconds for VLM snapshot triage. Offline training/planning tolerates higher latency

Compute levels and latency reflect *inference* on optimized hardware; training is higher and performed offline. p95 = 95th-percentile latency; DT = Digital Twin; RAG = retrieval-augmented generation; VRAM = video RAM; VLM = vision–language model; ROI = Return on Investment; SOP = Standard Operating Procedure *. Examples like Sora/Gen-2 are proprietary and may have access constraints

edge/cloud architectures that balance latency, privacy, and compute; and (iv) continuous evaluation, factuality, hallucination rate, privacy/bias tests tied to ITS KPIs such as time-to-awareness, mean time to repair (MTTR), and emissions reduction Sánchez-Navalón et al. (2024); Sznajder et al. (2025).

### 3.2 AIGC supporting technologies for ITS

AIGC-driven HMI in ITS depends on a coordinated stack that provides perception, compute placement, connectivity, simulation, sensing, and trust. At the perceptual layer, computer vision converts raw visual streams from CCTV, dashcams, and roadside units into actionable cues, incident detection, actor tracking, optical character recognition (OCR), and traffic-state estimates that condition and trigger generative behaviors such as explanatory alerts (text/TTS), scene reconstructions, and synthetic rare-event examples for training Dilek and Dener (2023); Lin et al. (2024c); Iglesias et al. (2023); Xu et al. (2025a). When embedded in HMI, these vision outputs also support evidence-cited summaries and human-interpretable rationales, improving transparency and operator trust in safety-critical workflows Adel (2023). Complementing perception, the Internet of Things (IoT) streams environmental, vehicular, and infrastructure telemetry that grounds prompts and closes actuator loops, enabling personalized advisories, predictive maintenance, and responsive control Zhu et al. (2025); Zhang et al. (2024d); Dou et al. (2023).

Edge–cloud placement and orchestration set the effective latency, privacy, and scalability envelopes for AIGC in ITS. Edge computing (vehicle/RSU/smart-intersection) brings redaction, feature extraction, and lightweight inference closer to data sources so operator-facing loops can meet few-seconds targets and privacy requirements; edges can also produce/transform outputs locally (e.g., emergency TTS) while reserving heavier multimodal generation for upstream services Xu et al. (2024a); Ye et al. (2024); Zeng et al. (2024); Xu et al. (2023a); Xie et al. (2024). Cloud platforms provide scale for training/fine-tuning, heavy multimodal inference, retrieval-grounding indices, policy/guardrail management, and media pipelines that back digital twins (DTs) Hu et al. (2024); Xu et al. (2024d); Wang et al. (2024b, 2023c); Panda et al. (2025). DTs mirror network state and host high-fidelity simulation and precomputed image/video libraries for drills, playback, and what-if analysis, thereby de-risking policies offline while grounding live recommendations when synchronized with field assets Ge and Qin (2024); Yang et al. (2024b); Irfan et al. (2024); Duran et al. (2025); Li et al. (2024b). In practice, edge excels on latency/privacy but is VRAM/thermal-limited; cloud maximizes scale/RAG (retrieval-augmented generation) freshness at WAN-latency/egress cost; DTs de-risk rare events offline but require simulation-to-reality synchronization; 5 G/6 G improves QoS (quality of service)/positioning yet hinges on slice policy and handover; IoT expands coverage while increasing device heterogeneity and missingness; and Distributed Ledger Technology (DLT) best fits asynchronous audit trails rather than inline control Cherukuri (2024).

Connectivity and trust complete the stack. 5 G/6 G contributes low-latency links, bandwidth, positioning, and slice-based QoS to backhaul vision frames and telemetry, deliver alerts, and synchronize edge–cloud workflows for autonomy/Advanced Driver Assistance Systems (ADAS) and distributed AIGC tasks; deployments are typically corridor- or district-scoped and require explicit QoS/SLA policies and robust handover Deng et al. (2023); Zhou et al. (2024b); Noor-A-Rahim et al. (2022); Jha et al. (2024); Trevlakis et al. (2023).

For provenance and auditability, blockchain notarizes prompts, model versions, retrieval sources, and generated outputs, and can enforce policy via smart contracts; given throughput/latency and privacy constraints, DLT is best applied to asynchronous audit trails and selective checkpoints, not inline control decisions that are better served by conventional data systems plus watermark/signature schemes Lin et al. (2023b); Luo et al. (2024a); Lin et al. (2024b); Liu et al. (2024c). Figure 6 summarizes these supporting technologies and their primary roles in AIGC-driven ITS.

In deployment, these technologies compose into a tiered edge–cloud pipeline: IoT sensors and computer vision provide the live context; edge nodes handle redaction, filtering, and low-latency text/TTS (and, when feasible, lightweight vision–language summarization on regions of interest); cloud services maintain the retrieval index and host heavier multimodal models; digital twins store precomputed image/video assets and support drills and playback; 5 G/6 G links carry control messages and bursty context between tiers; and blockchain/DLT notarizes prompts, model versions, retrieval sources, and outputs for audit. This division of labor keeps operator-in-the-loop interactions within a practical few-seconds target while shifting compute-intensive synthesis and validation to offline or precomputed paths, thereby balancing responsiveness, privacy, and governance in safety-critical ITS settings Jing et al. (2023); Gu et al. (2022); Nesen and Bhargava (2021); Avenoğlu et al. (2022).

Across the stack, trade-offs shape deployment choices: edge reduces latency and protects privacy but fragments compute/observability and must operate within VRAM and thermal constraints; cloud centralizes scale, RAG freshness, and governance but introduces Wide Area Network (WAN) latency and egress costs Hu et al. (2024); 5 G/6 G improves mobility-time connectivity and positioning yet demands careful QoS/slicing for uplink (video) and robust handover Deng et al. (2023); Jha et al. (2024); vision provides rich context but is sensitive to domain shift/drift, while digital twins mitigate operational risk by moving rare/haz-

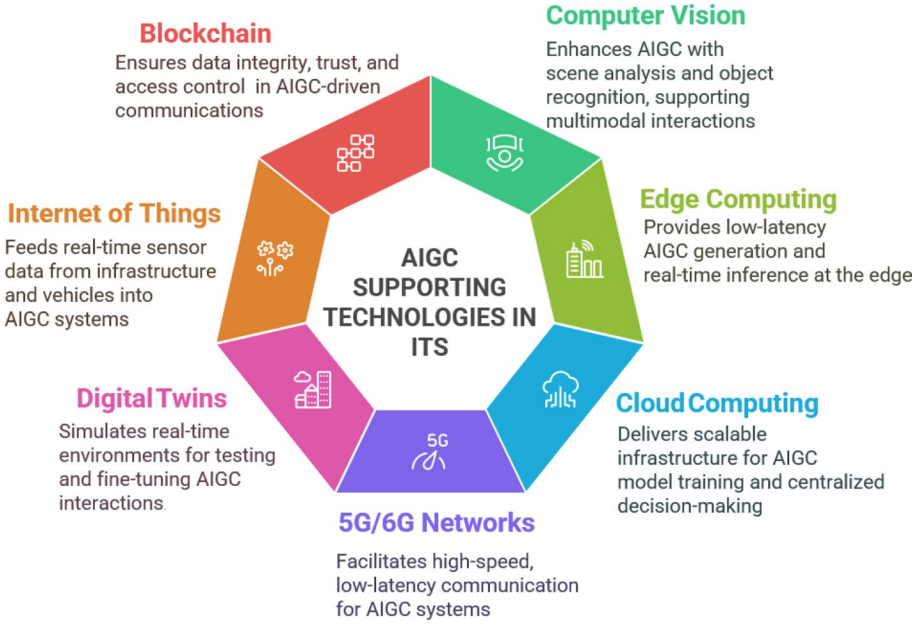



**Fig. 6** Overview of supporting technologies and their functional roles in AIGC-driven ITS

ardous cases into simulation at the cost of real–virtual synchronization and storage/compute pipelines Ge and Qin (2024); Duran et al. (2025); IoT expands coverage and personalization but adds device heterogeneity, data quality/missingness, and security/firmware management burdens; and blockchain brings provenance and tamper-evidence but remains throughput/latency limited; best suited to asynchronous audit trails and selective checkpoints rather than inline control-plane decisions Alharby (2023).

In terms of real-world feasibility, text/TTS at the edge is broadly production-ready for operator-in-the-loop tasks, and vision analytics with IoT telemetry is already operational when paired with redaction and monitoringRao et al. (2024); by contrast, heavy image/video generation is most practical as precomputed digital-twin assets served from edge caches, while online multimodal VLMs are feasible for snapshots or regions of interest when tiered (edge pre-processing plus cloud inference) and grounded via retrieval Wang et al. (2024c). Connectivity remains a gating factor: 5 G/6 G deployments are typically corridor- or district-limited, so service-level agreements should target specific control paths (e.g., alerts and advisories) rather than blanket guarantees, and bulk media or heavy inference should run asynchronously. Finally, blockchain adoption is most pragmatic when tied to audit and compliance, such as notarizing issued alerts or datasets, rather than serving as a general-purpose data plane Ampririt et al. (2022).

Put succinctly, our readiness view is: text/TTS at the edge is production; computer-vision plus IoT analytics is production (with redaction/monitoring); online image/video generation remains research/pilot with production use as precomputed DT assets; online VLMs for snapshot/ROI analysis (tiered edge→cloud) are pilot/limited operations; 5 G/6 G corridors are progressing from pilot to district-scale production; and DLT is production for provenance/audit but not for inline control.

Several open problems remain across orchestration, data, validation, assurance, sustainability, and standardization. First, edge–cloud orchestration requires more mature mechanisms for multi-model rollouts, such as versioning, canary testing, rollback, and consistent guardrails so that updates do not compromise safety or latency Pettinen and Hästbacka (2022). Second, retrieval-augmented generation (RAG) demands freshness guarantees and robust data governance across heterogeneous IoT and agency sources, including provenance tracking and access control Rodrigues and Branco (2024). Third, simulation-to-reality alignment for digital-twin assets requires continuous validation that improvements observed in simulation translate to perception/decision lift on held-out real data Luo et al. (2025). Fourth, privacy and safety guarantees for vision and multimodal pipelines must be engineered end-to-end, combining on-edge redaction with approval workflows for high-impact generations. Fifth, cost and carbon accounting for large-model operations is needed to keep deployments economically and environmentally sustainable at scale. Finally, standards and interoperability remain incomplete, spanning provenance (DLT), V2X QoS, and evaluation benchmarks that directly tie AIGC metrics to ITS KPIs, such as time-to-awareness, mean time to repair (MTTR), and emissions.

## 4 Applications of AIGC-driven HMI in ITS

This section examines the operational role of AIGC-driven HMI across five application domains: Smart Cities, Traffic Management, Safety & Security, Diagnostics, and Environment following the taxonomy in Sarwatt et al. (2024b) (see Fig. 7). For each domain, we articulate end-to-end workflows and decision points, then provide a comparative analysis along five axes: (i) latency targets and online/offline split; (ii) data prerequisites and grounding requirements; (iii) human-in-the-loop functions and escalation paths; (iv) deployment constraints with risk and assurance measures (privacy, drift, auditability); and (v) real-world feasibility/readiness (Production/Pilot/Research) with indicative KPIs such as time-to-awareness, spillback/delay reduction, detection precision/recall, MTTD/MTTR, and air-quality exceedance hours. This domain-centric framing emphasizes concrete operational trade-offs and feasibility within each domain, enabling practitioners to map capabilities to constraints and to quantify impact using domain-appropriate KPIs.

### 4.1 Smart cities

Smart cities use AIGC–driven HMI to improve day-to-day urban services such as mobility information, parking, advisories, while supporting longer-horizon planning and outreach. In practice, AIGC augments (rather than replaces) city staff: generative models draft and personalize messages, summarize conditions, and propose actions; human operators review, contextualize, and approve communications and policy-relevant changes. When coupled with IoT telemetry and open data, this division of labor enables more responsive, equitable, and transparent services Butt (2021).

Within smart parking, multimodal pipelines pair vision and IoT signals with text generation and text-to-speech (TTS) for guidance and alerts, and rely on diffusion/GAN assets for precomputed visualizations used in outreach and "what-if" planning. Generative models fuse live occupancy with historical demand to produce bay-level guidance and short-hori-

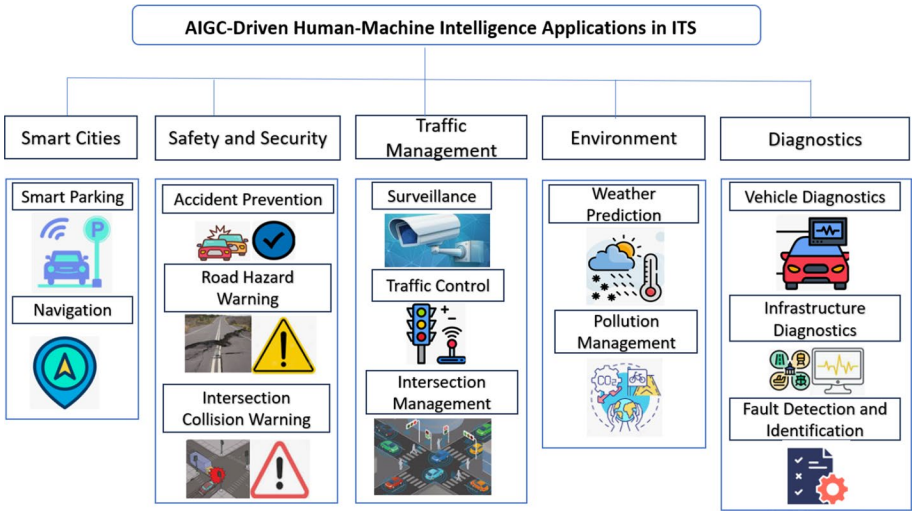



**Fig. 7** AIGC-driven HMI applications in ITS

zon forecasts; diffusion and semi-supervised autoencoder approaches increase robustness under sparse or noisy sensors Wei et al. (2023); Ge and Ma (2024). For navigation, AIGC systems synthesize incident-, weather-, and event-aware routing narratives and detours that adapt to evolving constraints, with route rationales surfaced for operators and travelers Niedoba et al. (2024); Matsumoto et al. (2024); Casale (2020). Recent work further personalizes trajectories by anchoring generated routes to real road graphs and nearby points of interest, improving usefulness and compliance over generic spatio-temporal baselines Lin et al. (2024a). Compared with smart parking, which tolerates minutes-scale forecast horizons and batch updates, navigation advice is latency-critical (seconds) and more sensitive to incident drift and handover gaps.

These capabilities involve clear trade-offs. Grounded text/audio reliably fits few-second latency budgets for citizen updates (e.g., detours, pricing changes) and operator advisories, and scales across languages; vision–language (VLM) snapshots add value when interpreting CCTV stills for incident context. By contrast, on-demand image/video synthesis is best reserved for offline digital-twin (DT) scenarios, public engagement, redesign mockups, and safety campaigns because online generation adds latency, compute cost, and governance overhead. Smart-city use leans more on public-facing communications (where accessibility and multilingual coverage dominate) and less on second-by-second control, so production value concentrates in retrieval-grounded text/audio and precomputed visuals rather than live visual synthesis Collins et al. (2021).

Deployment experience highlights recurring constraints. Data heterogeneity across sensors and vendors, data missingness, and domain shift (events, seasonal patterns) require continuous monitoring and freshness policies for retrieval-augmented generation (RAG). Privacy and personally identifiable information (PII) in camera feeds necessitate on-edge redaction before any generative step; pricing and enforcement introduce additional policy and ethics guardrails, including versioned prompts/models and auditable logs of issued advisories. Connectivity is corridor-dependent, and uplink budgets limit wholesale video transfer, pushing snapshot-oriented VLM use and edge pre-processing. Equity and accessibility remain first-order for citizen-facing outputs, multilingual TTS, plain-language text, and Americans with Disabilities Act (ADA)-compliant formats, supplemented by fallback channels (Short Message Service(SMS)/ Interactive Voice Response (IVR), kiosks) to mitigate digital-divide effects. Primary KPIs include time-to-park, cruising miles, Estimated Time of Arrival (ETA) error, advisory uptake, multilingual coverage, and complaint rate; while secondary KPIs include emissions/PM peaks near hotspots Zhang (2025).

From a feasibility standpoint, the most mature path today is to deploy retrieval-grounded text/audio advisories and occupancy summaries that tie directly to existing SOPs and live telemetry, these are production-ready. Snapshot-based VLM triage for parking anomalies and event-aware detours is viable under an approval-by-exception model and is best treated as pilot-stage functionality. By contrast, on-demand image/video synthesis for live citizen updates remains a research direction; operationally, it should be delivered as precomputed, digital-twin–backed media rather than generated online Sarker (2022).

### 4.2 Safety and security

Safety and security applications aim to prevent crashes, surface emerging hazards, and reduce high-risk interactions at intersections. In this domain, AIGC augments perception

and decision support rather than replacing operational control: generative models synthesize warnings, short-horizon forecasts, and operator summaries from camera feeds, V2X telemetry, weather, and incident logs, while human controllers validate and authorize interventions such as timing changes, lane closures, and public alerts.

In practice, vision-centric systems identify surface defects and hazards such as potholes, cracks, debris, or ice and trigger text or audio alerts to drivers and field crews Maeda et al. (2021). Diffusion and GAN simulators model rare, high-consequence events for AV training and tabletop exercises, expanding coverage to edge cases that are underrepresented in historical data Mishra et al. (2023). For crash risk, hybrid generative predictors (e.g., VAE and GAN) capture non-linear traffic patterns and improve short-horizon nowcasts Zhu et al. (2024), while large language models mine free-text incident reports to add contextual factors (lighting, work zones, human behaviors) that are often missing from structured fields Yoo et al. (2024). At conflict points, trajectory-aware systems leverage V2X to anticipate intersection collisions and recommend signal or advisory changes; generative components produce operator-readable rationales and driver-facing messages Meftah et al. (2024).

These capabilities come with material trade-offs. Snapshot-level vision–language assistance can improve situational awareness but adds GPU memory demand and seconds-scale latency; in safety-critical loops this is acceptable only when bounded by approval-by-exception and clear escalation. Generative simulators (GAN/diffusion models) are most effective offline for training, drills, and public safety campaigns because on-demand image/video synthesis increases latency and validation burden. By contrast, grounded text/audio scales across languages and reliably fits few-seconds alert budgets, provided outputs are retrieval-grounded in current SOPs and sensor truth to avoid hallucinated advisories. Compared with smart-city communications, safety workflows tolerate far less factual error and impose stricter governance on any change to control logic (signals, advisories) Xiao et al. (2023); Veretennikov et al. (2022)

Deployment exposes additional constraints. Data quality and domain shift (night, rain/snow, occlusions) degrade detection and prediction, necessitating continuous monitoring, periodic re-calibration, and selective re-labeling. PII in imagery requires on-edge redaction before any generative step, and auditability (who changed what, when) drives versioning of prompts/models alongside comprehensive logs of issued warnings. Connectivity is corridor-dependent; limited uplink budgets constrain continuous video backhaul, pushing edge pre-processing and snapshot-oriented VLM use. Human factors are pivotal: message timing, phrasing, and multilingual TTS affect driver compliance and should be evaluated through controlled field experiments with local populations (e.g., randomized comparisons of message variants).

From a feasibility standpoint, several patterns are deployment-ready today. Production systems can pair edge-filtered defect/hazard flags with operator confirmation and disseminate grounded text/TTS safety advisories; V2X-backed intersection warnings with human approval are likewise viable Wang et al. (2022b). Pilots increasingly use snapshot triage by VLMs for CCTV-detected anomalies and combine crash-risk nowcasting with LLM-derived context from narrative reports Yoo et al. (2024). By contrast, online image/video synthesis for live warnings and fully automated intersection conflict resolution using generative reasoning remain research-stage; where visual media is needed operationally, digital-twin, precomputed clips are preferable to on-the-fly generation NVIDIA (2023).

The near-term implication is that most value will come from edge-filtered detection and grounded text/audio with approval-by-exception, measured against KPIs such as time-to-awareness, near-miss reduction, and intervention MTTR. A priority research gap is simulation-to-reality validation: demonstrating that generative scenario training measurably reduces real-world collision surrogates at intersections and along high-injury corridors without inducing unintended driver behavior.

### 4.3 Traffic management

Traffic management uses AIGC–driven HMI to optimize flow, reduce congestion, and improve safety in day-to-day operations. Recent work shows a shift from static rules to data-driven, predictive control that adapts to complex urban patterns Andreoni et al. (2024); Ren (2024). Advanced platforms increasingly combine real-time sensing with generative models to forecast demand, recommend interventions, and draft operator-facing summaries; human supervisors validate and authorize signal timing changes, lane-use policies, and public advisories under approval-by-exception. At network scale, production systems already deploy sequence models for congestion prediction (e.g., RNN/transformer-based services) to support planning and response INRIX (2024), while cloud–edge collaborations integrate analytics with field assets under an advanced intelligent traffic management system (AITMS) umbrella Muriuki et al. (2024).

Across surveillance, control, and intersections, AIGC plays distinct but complementary roles. For surveillance, vision models enhanced with generative techniques improve detection under difficult conditions: illumination-adjustable GANs raise vehicle detection accuracy at night or in glare Zhou et al. (2023), and diffusion-based pipelines strengthen anomaly detection and yield timely flow information from video Wang et al. (2023a); Xue et al. (2024). For traffic control, text-to-traffic diffusion (e.g., ChatTraffic) turns natural-language scenarios into realistic microsimulations that help operators explore signal timing options before field deployment Zhang et al. (2024a); guided conditional diffusion supports near-real-time control policies that remain reviewable and tunable by staff Zhong et al. (2023). At intersections, knowledge-informed GANs model multi-vehicle trajectories to improve short-horizon forecasts Wei et al. (2024); spatio-temporal GANs extract patterns from sparse video Wang et al. (2022a); and autoencoder-based predictors anticipate pedestrian movement and identify high-risk interactions Du et al. (2023); Yu et al. (2024). In all cases, generative components supply “what-if” evidence and explanations, while human oversight ensures policy compliance and alignment with local context (schools, events, work zones).

These capabilities involve clear trade-offs. Snapshot vision–language assistance can speed anomaly triage but adds GPU-memory demand and seconds-scale latency, which must be bounded by operator-in-the-loop workflows. Generative simulators (diffusion/GAN) produce richer counterfactuals than classical tools and can reduce trial-and-error at the curb or signal cabinet, yet they introduce model risk and require calibration against field data. Sequence models for congestion forecasting scale well and meet minute-level horizons for corridor management, but their advantage over simpler baselines depends on retrieval-grounded inputs (incidents, weather, events) and continuous drift monitoring. Compared with safety applications, traffic management tolerates more offline computation (scenario search, timing design) but demands strong factual grounding for online messages that influence drivers in real time Xie et al. (2021).

Deployment experience highlights recurring constraints. Domain shift (night, rain, occlusion) and sensor missingness degrade surveillance and prediction, motivating periodic recalibration, selective re-labeling, and quality gates before any control change. Personally identifiable information (PII) in video requires on-edge redaction before any generative step; auditability drives versioning of prompts/models and comprehensive logs of recommended and enacted timings. Connectivity constraints limit continuous video backhaul, pushing edge pre-processing and snapshot-oriented vision–language use; cloud services host retrieval indices (procedures, events) and simulation backends with explicit change control. Human factors: operator load, explanation clarity, and multilingual citizen messaging directly influence compliance and should be evaluated through controlled field experiments with local populations Patrikar and Parate (2022).

From a feasibility standpoint, near-term deployments are strongest where forecasting and operator-in-the-loop control are coupled with retrieval-grounded summaries and digital-twin-tested timing plans. In practice, production today includes congestion prediction services and dashboard summaries for operators, grounded text/audio advisories tied to current SOPs, and surveillance flags with operator confirmation; pilots explore snapshot-based vision–language triage of CCTV anomalies and diffusion- or GAN-backed timing exploration with operator approval. Fully automated generative policy updates at signals and online image/video synthesis for live traveler information remain research-oriented. A priority research need is rigorous simulation-to-reality validation showing that generative scenario design and calibration deliver measurable reductions in delay and queue spillback on live corridors without adverse distributional impacts on vulnerable road users or adjacent streets.

### 4.4 Diagnostics

Diagnostics in ITS safeguard the dependable operation of vehicles and infrastructure by detecting emerging faults early, prioritizing maintenance, and shortening downtime. In AIGC–driven HMI settings, generative models synthesize realistic failure signatures, reconstruct missing signals, and summarize multi-sensor telemetry, while human experts validate findings, decide interventions, and schedule work orders. This collaboration spans three intertwined arenas: vehicle health, infrastructure condition, and fault detection and identification (FDI), which share data, models, and workflows.

Across vehicles, infrastructure, and system-level FDI, AIGC broadens coverage where ground-truth failures are sparse. On vehicles, GAN/autoencoder families generate or compress failure patterns to improve early-warning sensitivity and reduce false alarms; examples include MAGVA, a multi-hop attentive graph variational autoencoder that improves AV fault detection via realistic failure simulation Fu et al. (2023), and GAAD, a GAN-enabled autoencoder that flags sensor anomalies in real time for rapid recovery Rezaei et al. (2024). For civil assets such as bridges, tunnels, and pavements, generative models fuse IoT strain/gauge streams with imagery to detect incipient deterioration and to simulate progression under load and weather; GAN-based predictors have surfaced operational anomalies in large-scale IoT settings and supported crack/corrosion triage for timely maintenance Chen et al. (2022). At the system level, FDI pipelines combine variational autoencoders (VAEs) and simulation-augmented training: deep VAE classifiers can adapt to novel faults He and Jin (2021), while synthetic fault data produced with GANs improves gear/rotor diagnosis when real failures are rare Gao et al. (2021). In all cases, AIGC produces hypotheses

with confidence summaries, and maintenance engineers adjudicate actions (inspect, defer, replace) based on cost, risk, and service constraints.

There are clear trade-offs relative to classical diagnostics. Generative methods raise recall on weak signals and under missingness but introduce model risk and opacity. Compared with thresholds or the AutoRegressive Integrated Moving Average (ARIMA)-style baselines, VAEs and GANs often detect earlier but may overfit spurious correlations without strong regularization and drift monitoring. Graph VAEs (e.g., MAGVA) excel on fleet-scale, relational telemetry yet require careful graph construction and refresh cadence; GAN-augmented autoencoders (e.g., GAAD) heighten sensitivity to subtle sensor degradations at the cost of tuning stability. For infrastructure, image-conditioned diffusion/GAN pipelines detect surface defects under challenging lighting and weather better than static vision models, but they demand periodic recalibration to local materials and camera geometry. Overall, gains in early detection must be weighed against explainability, the potential to increase false-positive workload for crews, and the overhead of continuous validation.

Deployment in the wild adds practical constraints. Real-world telemetry is noisy and non-stationary: seasonal effects, software updates, and sensor aging shift distributions, necessitating continuous monitoring, scheduled re-labeling, and disciplined change control on models and prompts. Personally identifiable information (PII) in roadway imagery requires on-edge redaction before any generative step; for fleet data, tenancy/isolation and secure key management are mandatory. Connectivity limits (e.g., uplink budgets from depots or wayside units) favor edge pre-processing and batch uploads over continuous video backhaul; retrieval-grounded summaries (standard operating procedures, historical work orders) reduce hallucination in text outputs and align recommendations with current policy. Just as importantly, work-management integration matters: triage queues, service-level timers, and parts availability should condition model-suggested actions to avoid “ideal but infeasible” prescriptions Xu et al. (2023b).

In terms of feasibility and readiness, production deployments today commonly include anomaly screening with autoencoders/VAEs, grounded text summaries that populate work orders, and rule-constrained advisories to operators. Pilots increasingly use GAN/diffusion augmentation for rare-fault uplift, graph VAEs for fleet interactions, and image–IoT fusion for civil assets. Active research directions include automated generative root-cause narratives and learning loops that schedule inspections optimally across fleets and corridors Xu et al. (2025b). Digital twins are most effective as precomputed libraries of failure scenarios for drills and procedure rehearsal rather than as online media generators.

The near-term implication is that value concentrates in early detection paired with operator-in-the-loop triage, supported by retrieval grounding and audit trails, and tracked against KPIs such as mean time to detect (MTTD), mean time to repair (MTTR), repeat-fault rate, and cost per avoided failure. A priority research gap is rigorous field validation that synthetic-fault augmentation and generative reconstruction translate into fewer road calls and lane closures i.e., measurable gains in precision/recall at the deployed threshold, lower false alarms per hour, increased lead time, and improvements in maintenance throughput and safety outcomes (reduced MTTR, truck rolls, and lane-closure hours), rather than only higher offline discrimination metrics

### 4.5 Environment

Environmental conditions, especially weather and air quality, directly shape safety and efficiency across transport networks. In practice, AIGC augments environmental management by synthesizing short-horizon forecasts, imputing missing sensor data, and generating clear guidance for both operators and the public, while human supervisors validate outputs and decide interventions (e.g., speed advisories, lane restrictions, temporary low-emission zones). For weather, generative models improve nowcasting by learning distributions over historical and live measurements: GANs have enhanced ensemble forecasts and corrected rainfall biases, and language models translate complex meteorology into operational narratives suitable for traffic control and traveler information Bihlo (2021); Jeong and Yi (2023); Zhu et al. (2023). For air quality, hybrid architectures (e.g., Transformer–GAN and GAN–Neural Turing Machine) capture spatiotemporal pollution dynamics and likely sources, enabling targeted alerts and mitigation; traffic–energy co-models combining BiLSTM/CNN with GAN components link signal control decisions to emissions goals Zhao et al. (2024); Asaei-Moamam et al. (2023); Wan and Liu (2023).

ITS depends heavily on the environment because AIGC-driven HMI solutions enhance both weather prediction accuracy and pollution management. AIGC technologies use real-time data and predictive models with adaptive strategies to create smarter, sustainable environmental management systems that enhance transportation network safety and efficiency and environmental sustainability. Key applications are weather prediction and pollution management.

From an HMI perspective, the prevailing pattern is multimodal sensing and modeling upstream, with retrieval-grounded text or text-to-speech downstream. Weather and air-quality feeds from IoT stations, mobile sensors, and satellites drive generative predictors and scenario simulators. Their outputs are routed through templates aligned with local standard operating procedures (e.g., chain controls, speed harmonization, reversible lanes) to produce concise, multilingual advisories for operators and road users. Where appropriate, digital twins host precomputed visualizations (e.g., storm surge extents, wildfire smoke plumes) for planning and public outreach; day-of operations emphasize low-latency text/TTS and map overlays rather than on-demand image/video synthesis Chen et al. (2025).

Task-specific trade-offs are pronounced. Generative weather correction can reduce bias and sharpen spatial detail, but it introduces model risk and requires continuous verification against trusted gauges and radar; in regimes with abrupt shifts, simpler baselines may be more robust. Air-quality predictors that integrate traffic, meteorology, and land use improve hotspot anticipation yet remain sensitive to sensor siting, coverage gaps, and data drift Ton-Thien et al. (2023). Compared with traffic management, environmental applications tolerate more offline computation (scenario planning, public campaigns) but require especially high factual accuracy in public messaging, which favors grounded text/audio over live visual synthesis. Overall, AIGC adds the most value when it tightens short-horizon forecasts and reliably converts them into plain-language guidance tied to explicit actions.

Deployment reveals familiar constraints. Data quality and coverage vary (missing sensors, calibration drift), so pipelines need monitoring, periodic recalibration, and principled gap-filling before any generative step. Personally identifiable information in roadside imagery must be redacted at the edge; model and dataset versions for air-quality advisories require auditable records. Connectivity is corridor-dependent, uplink budgets discourage

continuous video backhaul, so snapshot-oriented vision and edge pre-processing are preferred, with cloud services maintaining retrieval indices (SOPs, ordinances) and running heavier simulations under change control. Equity and accessibility are first-order: messages should be multilingual, plain-language, and available over fallback channels (SMS/IVR, roadside signs) to mitigate digital-divide effects.

In terms of readiness, production deployments already include grounded text/audio advisories for weather and air-quality alerts, augmented by anomaly screening and bias-corrected forecasts that operators approve by exception. Pilot efforts span GAN/diffusion post-processing for precipitation nowcasts and hybrid pollution predictors that couple traffic control with emissions targets. Research priorities include retrieval-grounded multimodal explainers that link forecasts to specific control actions and rigorous field trials that quantify emissions and safety benefits. Near-term evaluation should track KPIs such as weather-related incident rates, delay and cancellation reductions, air-quality exceedance hours, population exposure, and distributional impacts, verifying not only model scores but measurable improvements for travelers and neighborhoods.

## 5 Existing case studies and initiatives of AIGC-driven HMI in ITS

This section presents important case studies and initiatives that demonstrate how AIGC-driven HMI integrates with ITS. The following subsections present the AIGC-driven HMIs' impacts on ITS performance enhancement.

### 5.1 Existing case studies

#### 5.1.1 51WORLD scenario copilot

51WORLD's Scenario Copilot pairs AIGC technologies, such as GANs and diffusion models with high-fidelity digital twins to generate rare, safety-critical driving situations from natural-language prompts, reducing coding burden and broadening scenario coverage for AV development Medium (2023). Integrated with detailed 3D city models (e.g., Shanghai's Pudong), the tool synthesizes snow, glare, and complex intersection scenes and interactions with highway structures as illustrated in Fig. 8 to stress-test perception and planning under controlled conditions. Reported benefits include shorter development cycles, lower test cost, and greater accessibility for smaller firms, supported by cloud-native, distributed simulation and human-in-the-loop review. Practical constraints remain: validating scenario realism against field data, fitting into incumbent toolchains, and enforcing data-privacy policies when mixing synthetic and real assets. Near-term roadmap items (NeRF-based rendering; adaptive traffic agents) target higher visual fidelity and behavioral realism; global expansion (Europe and North America) will hinge on governance (model/prompt versioning, audit trails) and integration playbooks Medium (2023).

#### 5.1.2 NVIDIA STRIVE/DRIVE sim

NVIDIA's STRIVE (Stress-Test Drive) uses adversarial optimization to auto-generate near-collision scenarios that probe AV planners inside the DRIVE Sim environment, then

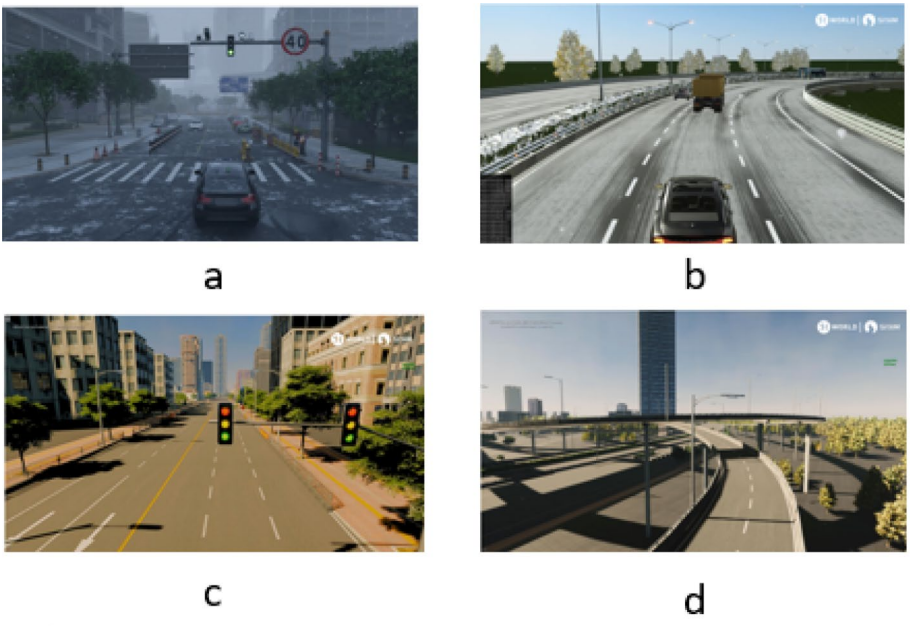



**Fig. 8** Various autonomous driving simulation testing scenarios by scenario copilot. **a** Under heavy snowy road conditions with a green traffic light and vehicles at an intersection, **b** snowy highway with a vehicle following a truck on a curved path, **c** clear urban street with green traffic lights at an intersection. **d** Highway overpass featuring urban infrastructure, testing the vehicle's interaction with highway structures Medium (2023)

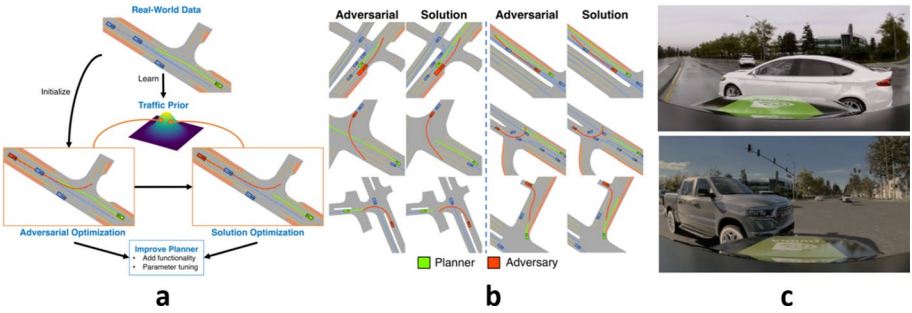



**Fig. 9** NVIDIA's STRIVE simulation workflow. **a** STRIVE generates traffic scenarios where an AV is involved in a near-collision, allowing developers to test and improve the decision-making processes of AV planners in simulated environments, **b** STRIVE not only generates these accidents but also provides potential solutions by applying generative models to optimize AV responses, **c** NVIDIA DRIVE Sim creates 3D simulations of potential accidents, which are vital for high-fidelity training in complex, real-world traffic conditions Rempe and Litany (2023)

explores counterfactual responses to harden policies under controlled but realistic conditions Rempe and Litany (2023). As illustrated in Fig. 9a, b, the pipeline both creates challenging incidents and proposes remedial maneuvers, enabling rapid, human-in-the-loop review of planner choices. DRIVE Sim further integrates AI-generated, data-driven

2D/3D accident models (Fig. 9c) to expand coverage of low-frequency, high-impact events without relying on real crash data, improving the diversity and repeatability of tests Hennessy (2022). Reported benefits include broader scenario coverage, faster iteration, and auditable decision traces; practical constraints include GPU cost, sim-to-real validation of scenario realism, and change control for models/prompts across releases. In practice, the approach is production-ready for simulation (planner regression, policy vetting) and pilot for downstream operational use, with success best measured by scenario diversity and difficulty metrics, planner robustness gains (e.g., reduced collision/critical-intervention rates), and documented compliance with safety governance Rempe and Litany (2023); Hennessy (2022).

## 5.2 Ongoing initiatives

### 5.2.1 INRIX compass

INRIX Compass applies generative AI (via Amazon Bedrock) over a 50-petabyte mobility data lake; vehicle telematics, mobile signals, transit, and parking to turn natural-language queries into traffic insights and recommended actions INRIX (2023). Practitioners can ask, for example, "Why is this street dangerous for pedestrians?" and receive retrieval-grounded explanations plus suggested interventions (signal timing changes, curb management, multimodal options), as illustrated in Fig. 10. The companion interface, INRIX IQ Mission Control (Fig. 11), combines historical and real-time visualizations to support approval-by-exception workflows and rapid operator decisions. Reported strengths are reduced analysis time, unified cross-dataset reasoning, and policy-ready summaries; practical constraints include data provenance and freshness (agency/partner feeds), governance for AI-generated recommendations, and change control for prompts/models. In terms of maturity, Compass is production for analytics and operator decision support, with roadmap features targeting climate resilience and AV integration. Representative KPIs include time-to-insight, crash hot-spot recall/precision, travel time reliability, and emissions reductions attributable to recommended interventions INRIX (2023).




**Fig. 10** Traffic insights and recommendations: showing how INRIX compass provides actionable recommendations Youtube (2024)

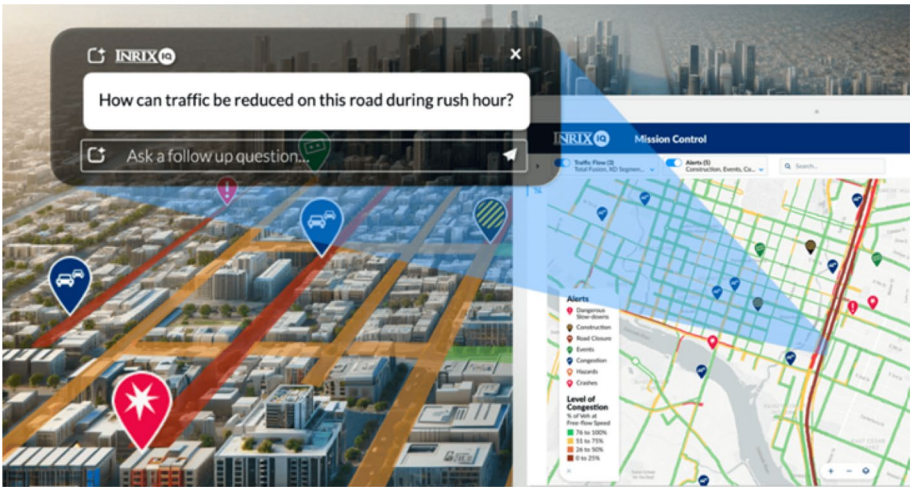



**Fig. 11** INRIX IQ mission control interface: displays real-time traffic data visualizations and user-friendly insights for decision-making INRIX (2023)

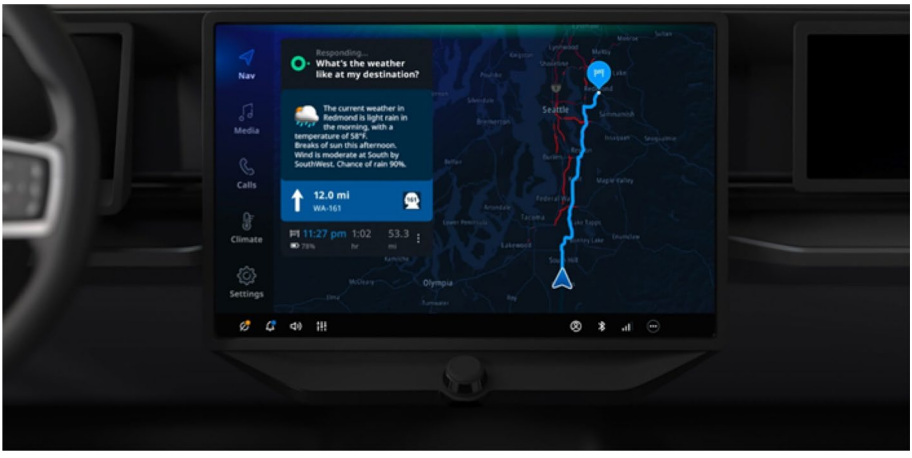



**Fig. 12** TOMTOM's AI-powered in-car navigation system using AIGC to generate real-time weather updates and personalized route information, enhancing driving decisions Daws (2023)

### 5.2.2 TOMTOM

TomTom, in partnership with Microsoft, leverages Azure's AI stack to synthesize driving scenarios from its HD Map corpus for AV training and hazard rehearsal, and to power an in-vehicle conversational assistant that provides voice navigation with real-time weather/route updates (Fig. 12). Beyond the cockpit, TomTom's ChatGPT-based planning integrates generative dialogue with live geolocation to optimize multi-modal trips, while generative tooling accelerates map-making and updates for traffic management and AV stacks. Observed benefits include faster scenario coverage, reduced map update latency, and hands-free HMI

that improves accessibility; operational constraints include privacy/PII handling in query logs, hallucination/grounding risks in conversational guidance, latency/QoS under variable connectivity, and governance for AI-suggested route changes. Readiness is production for voice/navigation and map maintenance, with pilot use of synthetic scenarios tied to AV development pipelines. Representative KPIs: guidance compliance rate, ETA error, re-route success rate under incidents/weather, and map freshness (update lag) attributable to the generative toolchain Daws (2023).

## 6 Evaluation framework for AIGC-driven HMI in ITS

While the development of AIGC-driven HMI systems for ITS is advancing, the evaluation frameworks for assessing their real-world effectiveness and operational performance are still in the early stages. To date, the literature has predominantly concentrated on single case studies or simulation-based experiments, which constrained understanding of the full-scale implementation of such systems in real world settings. In view of this, we present an overarching conceptual evaluation framework towards future research and practical implementation.

The assessment of AIGC-driven HMI system in ITS should be under a multi-layered framework, which concerns not only the performances of generative models, but also the quality of Human-Machine interaction, explainability, ethics compliance and deployment resilience. Since these real-time systems are becoming more dominant in determining the best transportation decisions and the execution of those decisions, their validation must reach beyond the accuracy of traditional models to consider human factors, societal trust, and operational status. The proposed framework is designed along five core layers: Functional (System Intelligence), Human-Centered Interaction, Explainability and Transparency, Ethical/Legal Compliance, and Robustness and Adaptability layers (See Fig. 13). Each layer is characterized by certain KPIs and the corresponding evaluation methods, allowing for evaluating reliability and the performance of the AIGC-driven HMI system in the context of ITS applications.

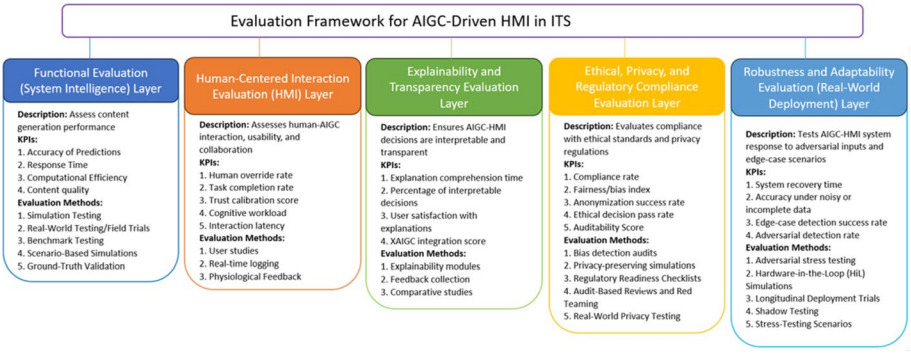



**Fig. 13** Five-layer evaluation framework for AIGC-HMI systems in ITS

### 6.1 Functional evaluation (system intelligence) layer

This layer evaluates the basic operating performance of AIGC-driven HMI systems applied in ITS domain. The performance evaluation measures the model precision as well as the response time, computational cost, and the quality of the content in order to evaluate the system in the ITS application including traffic management, incident prediction, and AV control.

The key performance indicators (KPIs) used to evaluate this layer include the following:

1. Accuracy of Predictions: How closely AI-generated predictions, such as traffic congestion and vehicle routes, align with real-world data or human expectations Zhang et al. (2024b).
2. Response Time: The time of processing data and making decisions that regulate traffic signals and re-route vehicles in AIGC system. Short response time is essential for real-time ITS applications, such as AV control and traffic management warning. Multimodal (2024).
3. Computational Efficiency: The amount of computational resources required for the system to generate real-time decisions.
4. Content quality: Utilizing metrics such as BLEU and ROUGE scores for text Papineni et al. (2002); Lin (2004), FID (Fréchet Inception Distance) for image quality Heusel et al. (2017), and SSIM (Structural Similarity Index Measure) for video and scene fidelity Wang et al. (2004). These metrics report semantic relevance and visual coherence of generative outputs.

Evaluation methods include:

1. Simulation Testing: Apply tools such as 51 WORLD's Scenario Copilot Medium (2023) and NVIDIA Corporation (2021)to regenerate the synthetic traffic scenarios to test the AIGC-driven HMI system in a laboratory setting. Such simulations allow system's performance evaluation under controlled environment prior to real use. Such as to test how an AV responds to extreme weather like blizzards and erratic traffic situations on a simulator.
2. Real-World Testing/Field Trials: Deploy the system in real environments to operate on native traffic management system data and AV test data. This approach provides an essential verification mechanism for evaluating the system in dynamic real operational environment and complex scenario. For example, prediction of traffic congestion by a traffic management system or an AVs response to unexpected obstacles in road during real world experiences.
3. Benchmark Testing: Assesses the performance of a system relative to industry standards and best practices, as well as competing products, to establish the level of efficiency, accuracy that the system could achieve in particular ITS tasks according to Zhang et al. (2024e). For instance, performance evaluation of AIGC-driven HMI traffic management system versus traditional traffic management systemin traffic rerouting and signal optimization.
4. Scenario-Based Simulations: Exercises the system under eccentric or complicated traffic situations that are vital for ITS, such as emergency response and large scale

disturbance. For instance, an accident scenario is simulated in AVs to evaluate system response to abrupt stops, detour routing, or emergency vehicle coordination

5. Ground-Truth Validation: Includes comparing outputs generated by AI with real-world data to confirm their accuracy and alignment with real-world situations. Such as validating a traffic prediction model by contrasting the prediction of the AI-generated with the real traffic flow data obtained from sensors or videos of traffic.

The assessment of these working parameters ensures that AIGC-driven HMI systems can produce real-time, context-aware, and scalable results, which are very important to the execution of critical ITS services. The technical basis of this layer allows the development of higher-level assessment layers including human-centered interaction and ethical compliance

### 6.2 Human-centered interaction evaluation (HMI) layer

The HMI layer assesses the performance AIGC in related HMI systems to assist human operators in ITS. This layer confirms the technical soundness while ensuring friendly user and reliable operation that conforms to human cognitive manner Tan et al. (2021); Sperrle et al. (2021). The AIGC-driven HMI system has to enable human operators to make decisions with good interaction facilities, such as good intuitiveness and awareness of context, in particular, in dynamic real-time ITS environment.

Key performance indicators (KPIs):

1. Human Override Rate: Measures how often human operators intervene or override AI-generated decisions. A higher override rate could indicate that these decisions are unclear, misaligned, or not trusted by humans.
2. Task Completion Rate: Measures how effectively human operators complete tasks with AIGC assistance.
3. Trust Calibration Score: Assesses the alignment between AIGC confidence levels and human trust in system decisions. It indicates whether the system fosters trust through transparent, explainable decision-making.
4. Cognitive Workload: Assesses the mental effort required from human operators during interaction with the AIGC system, often measured using tools like the NASA Task Load Index (NASA-TLX) Hart and Staveland (1988).
5. Interaction latency: Measures the time delay between system output (e.g., AI-generated suggestions) and human response.

To evaluate the human-centered interaction in AIGC-driven HMI systems, the following methods can be employed:

1. User Studies: involves manual work comparing with that supported by AIGC. It enables to compare the gains in efficiency and satisfaction while AI supports the decision-making Chand et al. (2025). Example: Traffic managers evaluate if automated predictions and adjustments enhance decision speed and accuracy using an AIGC-driven HMI traffic management system.
2. Real-Time Logging: Monitors how human operators interact with the system during real-time decision-making, logging interventions, adjustments, and overrides. Example:

Tracking how often drivers override navigation suggestions from an AV's AIGC-driven HMI system.

3. Physiological Feedback: Uses technology as eye-tracking, stress level sensors to track engagement of the user and cognitive load while interacting. This allows to gain insights into the mental workload associated with the operator to operate the system effectively. Example: Monitoring drivers' eye movements and stress levels while interacting with autonomous driving systems to ensure that the interface does not overwhelm the operator Xu et al. (2025c).

This layer ensures that the integration of HMI enabled by AIGC can be realized with both operation automation and human trust and safety. In the absence of strong human-in-the-loop design and assessment, AIGC's advantages in ITS could be compromised by dissonance, overdependence, or uninterpretability at crucial involvement.

### 6.3 Explainability and transparency evaluation layer

The layer of evaluation for explainability and transparency ensures that AIGC-driven HMI systems provide a result of decisions that is explainable and justifiable, which is a strong requirement for safety-critical ITS applications Chen (2024). In these contexts, understanding the rationale of AIGC outputs to foster trust and enable informed decisions is salient MacVittie (2024).

The evaluation of AIGC-driven HMI systems' explainability and transparency requires assessment of four specific KPIs:

1. Explanation Comprehension Time: Measures the time human operators need to understand the reasoning behind AIGC decisions, including route changes and incident detection.
2. Percentage of Interpretable Decisions: Represents the proportion of AIGC decisions that can be explained in simple, understandable terms. A higher percentage indicates greater transparency and human accessibility.
3. User Satisfaction with Explanations: Assesses how satisfied users are with the clearness, usefulness, and understandability of AI-generated explanations.
4. XAIGC Integration Score: Measures the quality of explainable content produced by the system, including natural language justifications, visual aids (e.g., saliency maps, decision heatmaps), and simulation replays.

Evaluation methods may include:

1. Explainability Modules: Real-time tools that provide clear justifications for AI decisions SmythOS (2025), showing why actions are taken (e.g., adjusting speed or changing routes). Example: In AVs, real-time explanations show why the vehicle adjusts its speed or detours, based on real-time data (e.g., traffic, weather conditions, or sensor inputs), helping build trust.
2. Feedback Collection: Collect user feedback on explanations to refine communication and improve clarity. Example: After a route change suggestion, a feedback form asks the driver whether the explanation was clear and if they trust the AI's recommendation.

3. Comparative Studies: Contrast various explanation media (e.g., text, visual, audio) to evaluate which best improves comprehension and trust. Example: In trafic control, evaluate visual heatmaps, textual explanations, and vocal alerts of traffic re–direction and ask controllers to rate clarity and effectiveness.

The evaluation of explainability and transparency is key to the trust in AIGC-driven HMI systems in ITS. Lack of explanations and system transparency may result in the accurate AIGC-based HMI systems being rejected or misused. This assessment procedure requires maintainability for these systems, but ensures that they can be audited, and that the human expectations are met, which ultimately results in safe and reliable decisions within the operational domain of the transportation system.

### 6.4 Ethical, privacy, and regulatory compliance evaluation layer

This layer checks if AIGC-driven HMI systems comply with ethical principles, privacy policies (e.g., GDPR), and relevant industry rules. This evaluation of compliance is critical to establishing public trust and receiving regulatory sanction and responsible deployment in transportation systems.

The following KPIs are used to support ethical, privacy, and legal evaluation:

1. Compliance Rate with privacy regulations: Measures the AIGC-driven HMI's adherence to privacy regulations such as GDPR Voigt and Von dem Bussche (2017), ISO/IEC 22989 HolisticAI (2023), and other relevant industry standards for data security and privacy.
2. Fairness/Bias Index: Assesses AIGC decision-making bias to guarantee fair and equitable outcomes, particularly in traffic management and autonomous driving systems.
3. Anonymization Success Rate: Evaluates the effectiveness of anonymization methods in safeguarding personally identifiable information that appears in AIGC-generated outputs, including synthetic driving data and personalized traffic responses.
4. Ethical Decision Pass Rate: Evaluates the AIGC-driven HMI's ability to manage ethical challenges, including emergency decision-making, while maintaining ethical standards and guidelines.
5. Auditability Score: Evaluates the transparency of AIGC-driven HMI decisions and the ease with which AIGC-driven decisions can be audited for accountability purposes.

Evaluation methods:

1. Bias Detection Audits: Use AI fairness frameworks to assess and mitigate bias in AIGC-driven HMI systems, ensuring equitable outcomes in decision-making. The open-source toolkit IBM AI Fairness 360 serves as an example of bias detection in AI models, according to Bellamy et al. (2019). In AVs, it can be employed to audit the AI's decision-making processes, ensuring that the model makes fair decisions regarding emergency braking. This means verifying that the system does not prioritize the safety of one demographic group (e.g., passengers) over another (e.g., pedestrians) based on factors such as age, gender, or ethnicity.

2. Privacy-Preserving Simulations: Conduct privacy-based checks such as membership inference attacks, reidentification attacks for discovering privacy leaks in AIGC-driven HMI systems to guarantee data privacy and security and compliance Wang et al. (2023b). For instance, a membership inference test reveals whether the system leaks personal data by indicating whether an individual's data was included in the training of the AIGC model. In AVs, this may involve checking whether the system says whether vehicles' data, e.g., travel patterns, were part of the training set (a potential privacy violation).
3. Regulatory Readiness Checklists: This is based on NIST Möller (2023) and OECD AI Ercan and Akin (2024) checklists to ensure that AIGC systems comply with privacy, data security, and ethical considerations at national and international levels. For example, the NIST AI risk management framework checklist aids AVs in fulfilling GDPR data security and privacy obligations. This enables organizations to perform risk assessments in a systematic manner that results in transparent and secure operation of the AIGC model.
4. Audit-Based Reviews and Red Teaming: Conduct audits and red teaming to evaluate ethical and privacy compliance, identify vulnerabilities, and ensure decisions align with ethical and regulatory standards Yulianto et al. (2023). For example, in the case of AVs, red teams simulate attacks or attempt to manipulate the vehicle's AI decision-making, such as causing the AV to make unethical or unsafe decisions (e.g., choosing between prioritizing the safety of a pedestrian or the driver) Al-Azzawi et al. (2025).
5. Real-World Privacy Testing: Performs field trials to assess privacy and ethical compliance, which verifies that AIGC-driven HMI systems fulfill privacy regulations and ethical standards in actual operational settings. For example, real-world privacy testing during autonomous driving trials requires tracking data management practices for passenger information, including location and travel patterns, to verify GDPR and other privacy regulations compliance.

Reviewing ethical, privacy and regulatory compliance is necessary to guarantee AIGC-driven HMI systems function in an ethical manner, respect users' privacy, and comply with relevant laws. Such audit ensures systems do not only comply with legal requirements but also contributes to build public trust and acceptance by mitigating biases, privacy risks, or ethical concerns.

### 6.5 Robustness and adaptability evaluation (real-world deployment) layer

Robustness and adaptability evaluation layer inquires whether AIGC-driven HMI systems can effectively manage edge cases, unexpected inputs, and adversarial attacks. The AIGC-driven HMI is shown to be functional and robust against unforeseen events such as weather perturbations, traffic crashes, and sensor malfunction, all of which are necessary to trust and ensure the safety of the ITS.

Key performance indicators (KPIs) include:

1. System Recovery Time: Measures the speed at which the AIGC-driven HMI system recovers from faults, adversarial interference, or operational failures, ensuring it

remains functional and resilient. Fast recovery times are essential to maintaining uninterrupted ITS operations and safety.
2. Accuracy Under Noisy or Incomplete Data: Assesses how well the AIGC-driven HMI system functions when it encounters imperfect data, including missing information and corrupted or low-quality inputs.
3. Edge-Case Detection Success Rate: Measures how well AIGC-driven HMI system identifies and handles unusual occurrences, including weather irregularities, unexpected traffic events, and pedestrian movements. A high detection rate ensures preparedness for safety-critical scenarios.
4. Adversarial Detection Rate: Evaluates how well AIGC-driven HMI system detects and counteract adversarial attacks, including sensor spoofing, data poisoning, and input manipulations Sarwatt et al. (2024a).

Evaluation methods:

1. Adversarial Stress Testing: Introduces synthetic perturbations such as altered road signs, extreme weather conditions, or adversarial attacks) using CARLA or SUMO to assess how well the AIGC-driven HMI system handles sensor failures, misleading data, or malicious inputs Ramakrishna et al. (2022).
2. Hardware-in-the-Loop (HiL) Simulations: Validate AIGC-driven HMI behavior in real-time control loops with physical hardware and simulated environments to assess performance against non-predictable conditions.
3. Longitudinal Deployment Trials: Observe the AIGC-driven HMI's performance, robustness, and adaptability during real-world deployments over extended periods Pascale et al. (2025); Zhou et al. (2025).
4. Shadow Testing: Test AIGC-driven HMI systems in parallel with live ITS infrastructures, ensuring no influence on real-time operations, to assess system resilience and adaptability before deployment.
5. Stress-Testing Scenarios: Simulate real-world scenarios such as severe weather, traffic accidents, or unexpected road closures to assess the system's resilience and adaptability under high-stress conditions.

ITS needs the HMI system driven by AIGC, to be strong and adaptive to make it safe, efficient, and reliable. The system should also be resilient to ensure that it can withstand the effects of uncertainties in operation, variations in the environment, and adversarial manipulation while maintaining the public's trust and the system's effectiveness in the field.

Although the suggested framework is conceptual in nature, it enables the definition of uniform test methods for AIGC-driven HMI systems in ITS. The framework specifies the KPIs and the validation procedures needed to assess the performance and flexibility of such systems. Hence, it is a foundation to establish evaluation procedures in the future to guide the application, scaling, and real-world validation of AIGC-driven HMI technologies under the domains of traffic management, AV systems, and emergency response.

## 7 Challenges and solutions of AIGC-driven HMI in ITS

In this section, we systematically investigate the main problem and solutions when applying AIGC-driven HMI technology to the ITS. These challenges are grouped into five main areas as summarized in Table 4: Technological, Data and Privacy, Human and Organizational, Security and Robustness, and Legal and Ethical challenges.

### 7.1 Technological challenges

The integration of AIGC with the existing ITS system infrastructure is still challenging, as many legacy systems(e.g., traffic management centers, field controllers, vehicle communications) were not designed with interfaces that could be cleanly abstracted into contemporary models and data services, resulting in silos and fragile hand-offs Teef et al. (2024); Zhang et al. (2024c). The practical upshot is modular adapters and open standards: APIs, middleware, and message buses that allow for piecemeal integration without toppling legacy systems Shrimal (2024). Meanwhile, real-time also has stringent latency budgets for operator-in-the-loop acts like rerouting and signal changes, as any delay directly reduces value. To reduce the turnaround time, deployments move pre-processing and inference towards the edge, meanwhile, model compression and pruning are introduced to meet device constrains Amar and Sanju (2024); Naveen and Kounte (2024).

Computing and orchestration requirements increase as deployments scale from corridors to regions. Cloud-native architecture allows for training, fine tuning and bursty inference with elastic capacity, while performance is maintained in the face of variable load and heterogeneous hardware by dynamic load balancing and containerization Randhi and Bandarapu (2024); Banerjee (2024).

In practice, a tiered edge–cloud architecture is formed: latency-critical filtering, redaction, and lightweight inference reside on vehicles/RSUs; heavier multimodal tasks and data lifecycle management are in the cloud. This does provide some incremental integration with measurable latency improvements today, while preserving a path to city-scale operation.

### 7.2 Data and privacy challenges

The execution of AIGC–driven HMI in ITS necessitates sensitive stream processing (vehicle locations, driver behaviors, incident video) with rigid privacy constraints. Practical instantiations integrate federated learning with raw data residing on vehicles or edge nodes and only aggregated updates being shared, along with anonymization pipelines that strip out personally identifiable information (PII) to comply with regulations (e.g., GDPR) and to keep public trust Erel-Özçevik et al. (2024); Farayola et al. (2024). These approaches maintain utility while decreasing the risk of exposure, allowing collaborative model improvement without centralization of raw data.

Fairness is a related concern. AIGC models may propagate historical and geographic biases, affecting signal prioritization, routing, or enforcement in unfair ways. Using diverse training corpora that cover different populations, environment, and operating conditions can improve generalization and reduce exclusionary outputs. Systematic bias audits should assess results by demographic group and neighborhood to detect disparities, and curated,

**Table 4** Summary of challenges, solutions, and future directions in AIGC-driven HMI for ITS

| Category | Challenge | Description | Solution | Future direction |
|---|---|---|---|---|
| =Technological | Integration and compatibility | Legacy systems struggle to interface with AIGC tools, causing data silos | Modular AIGC architectures, open standards (APIs, interoperable protocols) | =Self-evolving neuro-quantum AIGC ecosystems |
| | Real-time data processing | Latency delays critical decisions (e.g., accident response) | Edge computing, optimized algorithms (pruned neural networks) | |
| | Scalability | Systems struggle to handle growing data/usage demands | Cloud-native infrastructure, dynamic resource allocation (load balancing) | |
| =Data and privacy | Data privacy | Sensitive data (e.g., user locations) risks exposure | Federated learning, data anonymization (PII removal) | =Privacy-preserving and decentralized synthetic data ecosystems |
| | Bias in AIGC models | Biased training data leads to unfair outcomes | Bias audits, diverse training data (geographic/demographic coverage) | |
| | Data quality | Noisy/inconsistent data reduces model reliability | Data preprocessing/cleaning, anomaly detection (statistical filters) | |
| | Data quantity | Insufficient data limits model generalization | Synthetic data generation, crowdsourced data (public contributions) | |
| =Human and organizational | Human-AI collaboration | Confusion over roles between humans and AI | Collaborative interfaces (dashboards), human-centered design | =Emotion-aware human-AI collaboration frameworks |
| | Explainability and transparency | Black-box models erode trust in AI decisions | Explainable AI (XAI), model interpretability tools (saliency maps) | |
| | Adoption and resistance | Skepticism toward AI replacing human roles | Training programs, pilot projects (small-scale trials) | |
| =Security and robustness | Adversarial attacks | Malicious inputs deceive AIGC models (e.g., fake sensor data) | Adversarial training, real-time attack detection | =Autonomous, self-healing AIGC systems with integrated threat intelligence |
| | Model robustness | Models fail under stress (e.g., sensor failures) | Redundancy (backup sensors), data augmentation (simulated weather) | |
| | Security vulnerabilities | Cyberattacks exploit system weaknesses | Multi-layer security (encryption), regular security audits | |
| | Hallucination in AIGC models | AI generates false outputs (e.g., phantom traffic jams) | Hallucination detection, human validation (manual checks) | |
| =Ethical and legal | Accountability and liability | Uncertainty over responsibility for AIGC errors | Clear liability frameworks, open-source initiatives | =Decentralized ethical governance and autonomous justice systems |
| | Transparency and trust | Opaque AIGC processes undermine public confidence | Public audits, open-source sharing (code/datasets) | |
| | Regulation and governance | Lack of standardized policies for AIGC in ITS | AIGC governance frameworks, government regulations | |

representative datasets can further mitigate them Wang et al. (2024a); Bang et al. (2024). Incorporating these checks into the model lifecycle is necessary for responsible public use.

Model reliability is subject to the underlying data quality and coverage. It is known that noisy, missing and nonstationary signals deteriorate the predictions and recommendations, and strong preprocessing (filtering, imputation, anomaly detection) is required before any generative procedure Chan et al. (2023); Song (2024)). In cases where real-world data are limited, especially for infrequent events, synthetic training data is useful: diffusion/GAN/autoencoder pipelines produce realistic edge cases for training support, while crowdsourcing (e.g. GPS traces, user reports) for coverage extension in the spatial and temporal domains Cheng et al. (2026); Tamayo-Urgilés et al. (2024); Carlosama-Perachimba et al. (2024)). Together, these practices maintain privacy, improve fairness, and raise data fitness for purpose in production ITS.

### 7.3 Human and organizational challenges

A central barrier to AIGC–HMI adoption in ITS is misaligned human–AI workflows and unclear role boundaries. Operations teams need interfaces that surface model suggestions alongside the context operators already use, not parallel tools that fragment attention. Collaborative dashboards that blend real-time recommendations with operator inputs, incident metadata, and constraints support "approve-by-exception" patterns and reduce handoff friction Mossbridge (2024). Human-centred design further tailors interaction modes e.g., voice commands for drivers and field staff to match situational constraints without increasing cognitive load Shin et al. (2024); Ramos et al. (2024).

Trust depends on interpretability. Many generative models behave as black boxes, so explainability must be integral to HMI: plain-language rationales tied to the data that triggered a recommendation, simplified decision paths, and visual artefacts (e.g., heatmaps of contributing congestion factors, interactive flow charts) help operators verify that outputs align with local policy and ground truth Fenoglio and Kazim (2024); Rane et al. (2024). Embedding these explanations directly in operator workflows enables quick challenge/correct cycles and creates auditable traces of why actions were taken.

Adoption also hinges on organizational change. Staff unfamiliarity and job-security concerns can stall deployment unless agencies invest in targeted training, realistic tabletop/field exercises, and staged rollouts that demonstrate net benefits in a controlled scope Patil et al. (2024); Xu et al. (2024c). Pilot programs confined to a corridor or district, with clear success metrics (e.g., incident response times, schedule adherence), build confidence before scale-out Aydin and Dağlı (2025). Framing AIGC as augmentation, accelerating triage and communication while humans retain authority, helps secure buy-in and establishes sustainable roles and escalation paths across shifts and agencies.

### 7.4 Security and robustness challenges

AIGC models in ITS must withstand adversarial pressure and operational uncertainty. On the threat side, falsified or manipulated inputs may lead to dangerous recommendations or interfere with control loops; thus training is robust and multi-sensor corroboration is enforced. Adversarial training aids in robustness by training the models on specifically crafted perturbations, while Dempster–Shafer fusion in adversarial deep learning can aid

in identification of and removing attacks from the evidence sources Sarwatt et al. (2024a); Nagarajan et al. (2024). Concurrently, anomaly detection at the speed of telemetry is monitoring for anomalous pattern of system output (e.g., shock queues abrupt, trajectories erratic) and activates countermeasures of sensor isolation and fail-safe operation to protect the system function Hossain et al. (2024b).

Robustness to natural variation is also important. Field deployments experience challenges in sensor outages, extreme weather, and sudden changes in demand; production systems are thus architected with redundancy (alternative sensors, fallback algorithms) so, for example, radar and connected-vehicle feeds sustain monitoring when cameras fail Boujezza and Boubakri (2024). Data augmentation with simulated scenarios (rain, fog, noise) increases coverage, enhances generalization and mitigates brittleness in edge cases Pitale et al. (2024). These three steps: redundancy, augmentation, and policy-driven fallbacks are how reliability is boosted without placing too much faith in any one modality.

Cybersecurity risks also add to the attack surface since AIGC is combined with human supervision and end-user equipment. Sensitive information (the locations, trajectories) and safety-critical commands (signals, AV guidance) require defense in depth: encrypted communications, hardened endpoints, strict access control, and network segmentation Wang et al. (2024d); Hayagreevan and Khamaru (2024); Yang et al. (2024a).Regular security auditing and patching mitigate new vulnerabilities in software and hardware alike, and so the resilience against new threats is kept Teo et al. (2024).

Ultimately, hallucination is still a failure mode for generative systems. With time constraints, operators might place too much trust in outputs that are fluent but wrong, resulting in erroneous detours or misprioritized responses Wang (2024). Practical mitigations include hallucination screens that flag implausible content, retrieval-grounded prompts, and "approval-by-exception" for safety critical actions; high-stakes recommendations (e.g., emergency rerouting or signal overrides) should be subject to human validation or secondary model agreement Sahoo et al. (2024). Post-event feedback is further integrated into adaptive learning loops, resulting in further reduction in recurrence and calibration improvement.

### 7.5 Ethical and legal challenges

Since AIGC-driven HMI takes on decisions of increasingly significant impact in ITS, such as autonomous navigation and signal optimization, as well as emergency routing, the accountability and liability issues have become the focus. When things go wrong (e.g., AV collisions or misrouted emergency services), accountability may be spread among developers, operators, agencies, and end users. Clear liability frameworks for distributing duties and remedies (e.g., manufacturer liability for software defects versus operator misuse) that reduce contention and align incentives are necessary; the European AI Liability Directive reflects an effort towards clarifying tort exposure for AI-enabled systems EuropeanCommission (2022).

Trust is based on transparency. All stakeholders (policymakers, operators, and the public) must be able to know why an AIGC system chose a route, modified a signal plan, or raised an alert. Some practical techniques include the use of open artifacts (such as code, datasets, and decision logic that can be audited independently), explainable AIGC methods that offer natural-language rationales and visual traces, and public third-party audits to assess fairness, compliance, and performanceCañas et al. (2024); Chaudhary (2024). These

procedures render recommendations disputable and increase acceptance without revealing confidential IP or security information.

Regulation and governance remain uneven across jurisdictions, creating uncertainty for deployment and oversight. Absent standardized policies, operators face fragmented requirements on data protection, model monitoring, and ethical safeguards, increasing integration cost and risk. Coordinated policy development among agencies, industry, and standards bodies should establish enforceable governance for AIGC in transportation, covering data handling, incident reporting, model versioning, and redress supported by auditable processes and periodic review to keep pace with technical change Dey and Bhaumik (2024); Kashefi et al. (2024); Wang and Wu (2024); Arnold et al. (2024). In practice, agencies can phase in governance via approval-by-exception, retrieval grounding to current SOPs, immutable audit logs of issued recommendations, and documented escalation paths.

## 8 Future research directions

Building on the gaps identified in Sect. 7, we outline five forward-looking thrusts for AIGC-driven HMI in ITS. The emphasis is on directions that couple technical novelty with evaluability and policy readiness so advances translate into operational impact rather than isolated prototypes.

**Self-evolving neuro–quantum AIGC ecosystems.** A promising avenue is the co-design of AIGC pipelines across neuromorphic hardware, accelerated compute (including early quantum explorations), and federated topologies to deliver low-latency, energy-aware inference without centralizing raw data Ajayan et al. (2022); Behura and Patra (2025); Zhang et al. (2023). Such ecosystems would adapt online to urban growth and traffic anomalies, using event-driven spiking substrates for edge perception and cloud accelerators for policy search, while federated coordination preserves privacy by sharing model updates rather than records. Key questions include partitioning/scheduling under strict latency/energy budgets and proving end-to-end stability when components learn continuously.

**Privacy-preserving and decentralized synthetic-data ecosystems.** To overcome data scarcity and sensitive-data barriers, future AIGC-powered ITS should generate auditable, privacy-preserving synthetic corpora via federated diffusion or related generators trained on encrypted or locally held data He et al. (2024). Community validation can be supported through lightweight ledger mechanisms and participatory governance to surface bias and drift in synthetic assets Meneguzzo et al. (2024). Adaptive privacy models (dynamic anonymization, bias-aware sampling) are needed to balance privacy, fairness, and utility in downstream tasks Qian et al. (2025). Success hinges on rigorous evidence that synthetic data improves safety-critical KPIs without leaking or amplifying bias.

**Emotion-aware human–AI collaboration frameworks.** In control rooms and field operations, AIGC should modulate autonomy and explanation style based on operator cognitive load and affect (e.g., eye tracking, voice stress), escalating or deferring control accordingly Zou et al. (2025). Retrieval-grounded, plain-language rationales can bridge the explainability gap; simulation and gamified rehearsal can build calibrated trust before live deployment. The research need is safe adaptive-autonomy policies (when to hand off; how to summarize uncertainty) with measurable effects on workload, decision quality, and time-to-action.

**Autonomous, self-healing AIGC with integrated threat intelligence.** Generative pipelines must detect distribution shift and adversarial manipulation, isolate faults, and recover rapidly with minimal human intervention. Directions include shared threat intel for model hardening, generative "immune" mechanisms to regenerate compromised components, and physics/constraints checks (including quantum-annealing-style validators) to bound hallucinations in safety-critical use Mario et al. (2024). Research priorities are red-team benchmarks tailored to AIGC-for-ITS, rollback/playbook automation, and quantitative resilience metrics (MTTD/MTTR) at the system level.

**Decentralized ethical governance and accountability.** To earn public trust, provenance and recourse must be first-class citizens. Lightweight, privacy-preserving attestations of prompts, training evidence, model versions, and issued advisories can support audits and dispute resolution; DAO-style mechanisms can involve stakeholders in policy evolution Saurabh et al. (2024). "Ethical digital twins" can simulate impacts and distributional effects before rollout, while contractual templates (smart contracts and conventional agreements) clarify liability and remediation. The agenda is to make governance actionable measurable audit coverage/delay, transparent appeal paths, and community-level fairness reporting without compromising operational latency.

## 9 Conclusion

This review synthesized how Artificial Intelligence–Generated Content (AIGC) combined with Human-Machine Interaction (HMI) can enhance Intelligent Transportation Systems (ITS) across smart-city services, traffic management, diagnostics, safety and security, and environment. On the core side, we analyzed five generative modalities: text, image, video, audio, and multimodal vision–language models (VLMs) clarifying roles, quality–latency trade-offs, and deployment readiness for operator-in-the-loop workflows (e.g., text/audio in production; image/video as precomputed assets; VLMs in pilot/limited operations). On the enabling side, we examined supporting technologies: computer vision, IoT, edge/cloud computing, digital twins, 5 G/6 G connectivity, and blockchain/DLT that feed, execute, validate, and audit generative pipelines. Industry case studies illustrated practical value in scenario simulation, real-time advisories, and human-centered decision support, while revealing integration, privacy, and governance constraints. To move beyond descriptive reporting, we proposed a five-layer evaluation framework: functional performance, human interaction, explainability, ethical/compliance, and robustness to align AIGC-driven HMI assessment with operational KPIs such as time-to-awareness, MTTR, incident precision/recall, and emissions/throughput deltas. We also surfaced open challenges around data privacy and PII handling, legacy and policy integration, bias and drift management, cost/carbon efficiency, standards, and simulation-to-reality validation. Looking ahead, we see the most immediate impact from retrieval-grounded text/audio and snapshot VLM triage under approval-by-exception, complemented by precomputed visual media in digital twins; future research should prioritize multimodal workflow refinement, freshness-aware RAG and model governance, rigorous field evaluation tied to ITS KPIs, and scalable, ethical deployments that keep humans in the loop where safety and accountability are paramount.

**Acknowledgements** This research was partly supported by the European Celtic+ and Swedish Vinnova project (C2022/1-3) "CISSAN-Collective Intelligence Supported by SecurityAware Nodes", and the Knowledge

Foundation, Sweden, through the Human-Centered Intelligent Realities (HINTS) Profile Project (contract 20220068).

**Author contributions** DSS: conceptualization, literature search and collection, methodology, validation, writing-original draft, writing-review and editing, visualization. FK, AAKR, AGP: literature search and editing. JD, HN: conceptualization, revision, and supervision of the manuscript; writing-original draft and editing.

**Funding** Open access funding provided by Blekinge Institute of Technology.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare no conflict of interest.

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**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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